

2.2 Explicitly Defined Population Attributes

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“Statistics is: the fun of finding patterns in data; the pleasure of making discoveries; the import of deep philosophical questions; the power to shed light on important decisions, and the ability to guide decisions. . . in business, science, government, medicine, and industry. . . ”

– Professor David Hand

2.2.1 Population Attributes

Overview

- We define what it means to be an attribute of a population and discuss important attributes such as location, spread and skewness.
 - We consider an example using an agriculture population of farms.
 - This population has missing values and we see how to deal with them in R.
-

- The population is typically a set or collection of units, each with one or more variates that we can measure.
- **Variates** are characteristics of each unit in the population, and they can take on numerical or categorical values.
 - The values of variates typically differ from unit to unit.
 - If we are only interested in the variate y 's we might write the population as

$$P = \{y_1, y_2, \dots, y_N\}$$

- **Population attributes** are summaries describing characteristics of the population.
 - Formally, an attribute is a **function** applied to the entire population and determined by the variate values observed for each of the population's units.

$$a(P) = f(y_1, y_2, \dots, y_N)$$

- Some examples of attributes are
 - the population total:

$$a(P) = \sum_{u \in P} y_u$$

- or various counts over the population

$$a(P) = \sum_{u \in P} I_A(y_u)$$

where $I_A(y)$ is the indicator function

$$I_A(y) = \begin{cases} 1 & \text{if } y \in A \\ 0 & \text{if } y \notin A \end{cases}$$

EX: # of humans taller than 6 feet

$$a(P) = \sum_{u \in P} I_A(y_u)$$

where y_u is a height in feet and A is all heights larger than 6 feet.

- Example: Let $P = \{1, 2, \dots, 100\}$ and calculate $a(P)$: the total number of units which are multiple of 10.

$$a(P) \text{ as above with } I_A(y) = \begin{cases} 1 & \text{if } y \bmod 10 = 0 \\ 0 & \text{if } y \bmod 10 \neq 0 \end{cases}$$

- In general, attributes can be numerical or graphical – as long as they summarize the whole population.

- A histogram of y_u values
- A scatter-plot of the (x_u, y_u) pairs
- The least squares estimate of the line-of-best-fit
- The residual variation around the line-of-best-fit

Location Attributes

These attributes measure or describe the centre of the distribution of variate values in a dataset.

- the population average:

$$a(P) = \bar{y} = \frac{1}{N} \sum_{u \in P} y_u$$

- the population proportion:

$$a(P) = \frac{1}{N} \sum_{u \in P} I_A(y_u)$$

- Other examples include the mode, the median, etc.

Spread Attributes

These attributes measure variability or spread of the variate values in a data set. Some are

- the population variance:

$$a(\mathcal{P}) = \frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \bar{y})^2$$

- the population standard deviation:

$$a(\mathcal{P}) = SD_{\mathcal{P}}(y) = \sqrt{\frac{\sum_{u \in \mathcal{P}} (y_u - \bar{y})^2}{N}}$$

- coefficient of variation:

$$a(\mathcal{P}) = \frac{SD_{\mathcal{P}}(y)}{\bar{y}}$$

- Note:** the population variance or standard deviation could also be defined using $N - 1$ in the denominator.
- Other examples are the range, the inter-quartile range, etc.

Order Statistics

$y_{(i)} = i^{\text{th}}$ smallest value in $\mathcal{P}$

- Population attributes can also be based on an indexed collection of values,

$$\text{min} \rightarrow y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(N)} \leftarrow \text{max}$$

which are the the variate values $y_u \in \mathcal{P}$ ordered from smallest to largest (including ties).

- This ordering can be informative, for example:

```
### read the data from wherever it is stored, e.g. in some directory named Data
setwd("/Users/nstevens/Dropbox/Teaching/STAT_341/Fall_2023/Lectures/Data/")
agpop <- read.csv("agpop_data.csv", header=TRUE)

par(mfrow=c(1,2))
y <- agpop$acres87[agpop$region == "NE"]
y = na.omit(y)

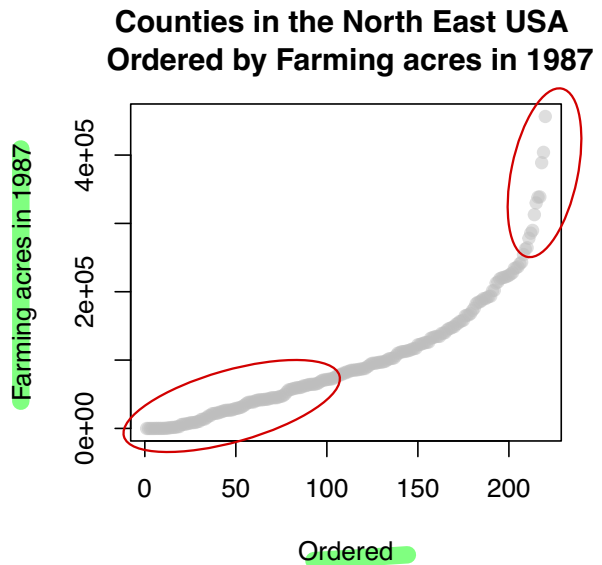
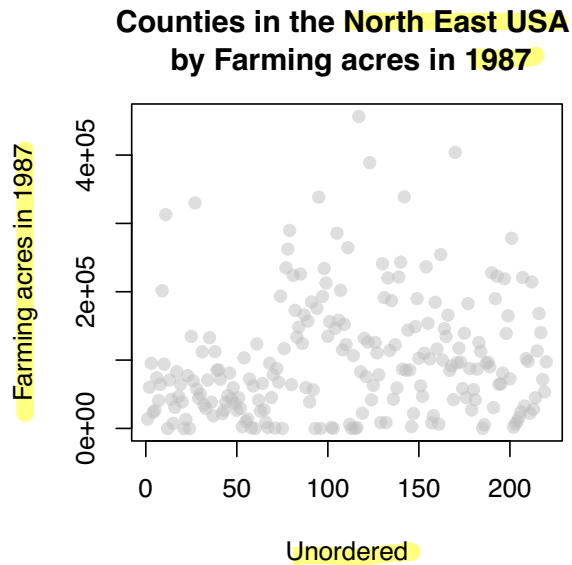
plot(y, pch = 19, col=adjustcolor("grey", alpha = 0.5),
     xlab = "Unordered",
     ylab = "Farming acres in 1987",
     main = "Counties in the North East USA \n by Farming acres in 1987")
```

```

yordered <- sort(y)
yrank <- rank(y, ties.method = "first") # Now ensure ties appear in data set order

plot(yrank, y, pch = 19, col=adjustcolor("grey", alpha = 0.5),
     xlab = "Ordered",
     ylab = "Farming acres in 1987",
     main = "Counties in the North East USA \n Ordered by Farming acres in 1987")

```



Location Attributes based on Order Statistics

- the population minimum:

$$a(P) = \min_{u \in P} y_u = y_{(1)}$$

- the population maximum:

$$a(P) = \max_{u \in P} y_u = y_{(N)}$$

- the population mid-range:

$$a(P) = \frac{1}{2} \left[\min_{u \in P} y_u + \max_{u \in P} y_u \right] = \frac{y_{(1)} + y_{(N)}}{2}$$

- the population median:

$$a(P) = \text{median } y_u = \begin{cases} y_{((N+1)/2)} & \text{if } N \text{ is odd} \\ \frac{y_{(N/2)} + y_{(N/2+1)}}{2} & \text{if } N \text{ is even} \end{cases}$$

- the population quartiles:

- Q_1 is 25th percentile, or the first quartile,
- Q_2 is 50th percentile, or the median, and
- Q_3 is 75th percentile, or the third quartile.

Variability Attributes based on Order Statistics

- The population range:

$$a(P) = \max_{u \in P} y_u - \min_{u \in P} y_u = y_{(n)} - y_{(1)}$$

- The population inter-quantile range IQR:

$$a(P) = Q_3 - Q_1$$

where Q_1 and Q_3 are 25th and 75th percentiles or the first and third quartiles, as above.

- The Median Absolute Deviation (MAD) is the median of the absolute differences between each y_u and the median:

$$a(P) = \text{median}_{u \in P} \left| y_u - \text{median}_{u \in P} y_u \right|$$

Skewness Attributes



These are measures of asymmetry in a population. A symmetric distribution of population values should result in a skewness attribute of zero.

- Pearson's moment coefficient of Skewness:

$$a(P) = \frac{\frac{1}{N} \sum_{u \in P} (y_u - \bar{y})^3}{[SD_P(y)]^3}$$

- Pearson's second skewness coefficient (median skewness) given by

$$a(P) = 3 \times \frac{(\bar{y} - \text{median}_{u \in P} y_u)}{SD_P(y)} \quad \star$$

- Bowley's measure of skewness based on the quartiles:

$$a(\mathcal{P}) = \frac{(Q_3 + Q_1)/2 - Q_2}{(Q_3 - Q_1)/2}$$



is an order statistic analogue to



Agriculture Data Example

- Recall the US Census of Agriculture population which measures the decline in farms from 2007 to 2012. The dataset represents a population of $N = 3,078$ units, where each unit is a county (or county equivalent) as defined by the US Census of Agriculture.

Read the data from wherever it is stored, e.g. in some directory named Data

```
directory <- "../..Data"
dirsep <- "/"
filename <- paste(directory, "agpop_data.csv", sep=dirsep)
agpop <- read.csv(filename, header=TRUE, stringsAsFactors=TRUE)
```

A number of population attributes can be calculated via `summary(agpop)`

`summary(agpop)`

category

##	county	state	acres92	acres87
##	WASHINGTON COUNTY: 30	TX : 254	Min. : -99	Min. : -99
##	JEFFERSON COUNTY : 25	GA : 159	1st Qu.: 80903	1st Qu.: 86236
##	FRANKLIN COUNTY : 24	KY : 120	Median : 191648	Median : 199864
##	JACKSON COUNTY : 23	MO : 114	Mean : 306677	Mean : 313016
##	LINCOLN COUNTY : 23	KS : 105	3rd Qu.: 366886	3rd Qu.: 372224
##	MADISON COUNTY : 19	IL : 102	Max. : 7229585	Max. : 7687460
##	(Other) : 2934	(Other) : 2224		
##	acres82	farms92	farms87	farms82
##	Min. : -99	Min. : 0.0	Min. : 0.0	Min. : 0.0
##	1st Qu.: 96397	1st Qu.: 295.0	1st Qu.: 318.5	1st Qu.: 345.0
##	Median : 207292	Median : 521.0	Median : 572.0	Median : 616.0
##	Mean : 320194	Mean : 625.5	Mean : 678.3	Mean : 728.1
##	3rd Qu.: 377065	3rd Qu.: 838.0	3rd Qu.: 921.0	3rd Qu.: 991.0
##	Max. : 7313958	Max. : 7021.0	Max. : 7590.0	Max. : 7394.0
##	largef92	largef87	largef82	smallf92
##	Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.00
##	1st Qu.: 8.00	1st Qu.: 8.00	1st Qu.: 8.00	1st Qu.: 13.00
##	Median : 30.00	Median : 27.00	Median : 25.00	Median : 29.00
##	Mean : 56.18	Mean : 54.86	Mean : 52.62	Mean : 54.09
##	3rd Qu.: 75.00	3rd Qu.: 70.00	3rd Qu.: 65.00	3rd Qu.: 59.00
##	Max. : 579.00	Max. : 596.00	Max. : 546.00	Max. : 4298.00
##	smallf87	smallf82	region	

```
## Min.    : 0.00   Min.    : 0.00   NC:1054
## 1st Qu.: 17.00   1st Qu.: 16.00   NE: 220
## Median : 35.00   Median : 34.00   S :1382
## Mean    : 59.54   Mean    : 60.97   W : 422
## 3rd Qu.: 67.00   3rd Qu.: 67.00
## Max.    :3654.00   Max.    :3522.00
##
```

- The first two, and last, variates (`county`, `state`, and `region`) are categorical
 - we need to handle them differently if we want to “summarize” them
- It is customary to count the number of each instance of a categorical variable’s levels:
 - For instance, the number of counties in each state, or the number of counties in each region:

```
summary(factor(agpop$state))
```

```
## AK AL AR AZ CA CO CT DE FL GA HI IA ID IL IN KS KY LA MA MD
## 5 67 75 15 58 63 8 3 67 159 4 99 44 102 92 105 120 64 14 23
## ME MI MN MO MS MT NC ND NE NH NJ NM NV NY OH OK OR PA RI SC
## 16 83 87 114 82 56 100 53 93 10 21 33 17 62 88 77 36 67 5 46
## SD TN TX UT VA VT WA WI WV WY
## 66 95 254 29 98 14 39 72 55 23
```

```
summary(factor(agpop$region))
```

```
## NC NE S W
## 1054 220 1382 422
```

- The remaining variates are numeric, so the summary provides
 - the average (`mean`)
 - the minimum (`min`) and maximum (`max`)
 - the first and third quartiles
 - the median (`median`)

```
summary(agpop[,-c(1,2,15)]])
```

```
##      acres92      acres87      acres82      farms92
## Min.    :   -99   Min.    :   -99   Min.    :   -99   Min.    :    0.0
## 1st Qu.: 80903   1st Qu.: 86236   1st Qu.: 96397   1st Qu.: 295.0
## Median :191648   Median :199864   Median :207292   Median : 521.0
## Mean    :306677   Mean    :313016   Mean    :320194   Mean    : 625.5
## 3rd Qu.:366886   3rd Qu.:372224   3rd Qu.:377065   3rd Qu.: 838.0
## Max.    :7229585   Max.    :7687460   Max.    :7313958   Max.    :7021.0
##      farms87      farms82      largef92      largef87
## Min.    :    0.0   Min.    :    0.0   Min.    :    0.00   Min.    :    0.00
## 1st Qu.: 318.5   1st Qu.: 345.0   1st Qu.:    8.00   1st Qu.:    8.00
## Median : 572.0   Median : 616.0   Median :   30.00   Median :   27.00
## Mean    : 678.3   Mean    : 728.1   Mean    :   56.18   Mean    :  54.86
## 3rd Qu.: 921.0   3rd Qu.: 991.0   3rd Qu.:   75.00   3rd Qu.:   70.00
```

```
## Max. :7590.0 Max. :7394.0 Max. :579.00 Max. :596.00
## largef82 smallf92 smallf87 smallf82
## Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.00
## 1st Qu.: 8.00 1st Qu.: 13.00 1st Qu.: 17.00 1st Qu.: 16.00
## Median : 25.00 Median : 29.00 Median : 35.00 Median : 34.00
## Mean : 52.62 Mean : 54.09 Mean : 59.54 Mean : 60.97
## 3rd Qu.: 65.00 3rd Qu.: 59.00 3rd Qu.: 67.00 3rd Qu.: 67.00
## Max. :546.00 Max. :4298.00 Max. :3654.00 Max. :3522.00
```

- Looking at the number of acres devoted to farms (i.e. `acres92`, `acres87`, `acres82`) reveals something curious
 - the minimum of each is -99 which is a strange value for the number of acres!
 - No acreage should be less than zero.
 - Missing data are encoded as -99 in this data set.
 - These should be replaced by NA which is the standard representation for missing data in R.

```
## acres92 acres87 acres82
## Min. : -99 Min. : -99 Min. : -99
## 1st Qu.: 80903 1st Qu.: 86236 1st Qu.: 96397
## Median : 191648 Median : 199864 Median : 207292
## Mean : 306677 Mean : 313016 Mean : 320194
## 3rd Qu.: 366886 3rd Qu.: 372224 3rd Qu.: 377065
## Max. :7229585 Max. :7687460 Max. :7313958
```

Having encoded the missing values as NA, the summary of these variates will now reflect the changes.

```
### which values are missing can be determined with a logical query
missing92 <- agpop[, "acres92"] == -99
### missing92 is a logical vector of the same length
### as agpop[, "acres92"] containing a TRUE in every
### position where a -99 appeared and FALSE everywhere else.
### The total number of missing values can be had by
### summing (because logical TRUE is treated as 1, and FALSE as 0)
sum(missing92)
```

```
## [1] 19
```

```
### Alternatively, the `which` function could be used
### to identify the row numbers
rowNumsMissing <- which(agpop[, "acres92"] == -99)

### The values can be changed to NA by using these locations
### (either rowNumsMissing or missing92) to identify the rows
### and replace the values
agpop[missing92, "acres92"] <- NA

### The same can be done for the other two acreages
agpop[agpop[, "acres87"] == -99, "acres87"] <- NA
agpop[agpop[, "acres82"] == -99, "acres82"] <- NA
```

```
summary(agpop[,c("acres92", "acres87", "acres82")])
```

```
##      acres92      acres87      acres82
## Min.   :    0 Min.   :    0 Min.   :    0
## 1st Qu.: 82446 1st Qu.: 87530 1st Qu.: 97835
## Median :193688 Median :201728 Median :209222
## Mean   :308582 Mean   :315374 Mean   :321973
## 3rd Qu.:368482 3rd Qu.:374576 3rd Qu.:379172
## Max.   :7229585 Max.   :7687460 Max.   :7313958
## NA's   :19 NA's   :23 NA's   :17
```

NAs in R

- **Note** that many programs in R accommodate missing data (represented as NAs) and do something appropriate (typically they omit them).
 - For your own code and analyses, you either need to decide what to do with NAs or ensure that the data do not have any NAs.
 - If you choose to simply omit NAs, for example, the function `na.omit(...)` may be helpful (it will remove rows which contain an NA from a data set). For other possibilities see `help("na.omit")` in R.

? na.omit

Summary

- We reviewed ways to numerically and graphically describe or summarize a population.
- Some attributes can describe the same feature of a population.
- Some attributes can describe different features of a population.

Exercise

- From the Explicit Attributes exercises try problem 24.

2.2.2 Attribute Properties

Overview

- We examine the behaviour of an attribute when we change the units in some way, or when we replicate the population.
-

- A population attribute is a function of measured variates y_u :

$$a(P) = f(y_1, y_2, \dots, y_n)$$

and the variates y_u are typically associated with some measurement units. In general we are interested in understanding how an attribute changes

- when we change the units of measurement, and
- when we change the population.

- Sometimes only the scale of measurement is changed:

distance
volume
weight

- 1 yard = 3 feet; 1 mile = 5280 feet; 1 metre = 1000 mm;
- 1 imperial gallon = 4.54609 litres; 1 US gallon = 0.832674 imperial gallon;
- 1 kilogram = 1000 grams = 2.20462 pounds.

- Sometimes only the location of the zero for that measurement is changed:

temperature

- absolute zero is 0° Kelvin or -273° Celsius
- 1 Celsius degree = 1 Kelvin degree (no change in scale of measurement)

- Sometimes both the location and the scale change:

- water freezes at 0° Celsius = 32° Fahrenheit (location change)
- 1 Celsius degree = 1.8 Fahrenheit degrees (scale change)

- Sometimes the change involves more than just a change in location and/or scale of measurement:

- fuel economy might be reported in miles per gallon (US or Imperial) or litres per hundred kilometres.
- the Richter scale for earthquakes is a logarithmic measure of the amplitude of seismic waves.

running speed: miles per hour
minutes per km

no change
↓commensurate
↙ change**Location Invariance and Equivariance**

For an attribute $a(\mathcal{P}) = a(y_1, \dots, y_N)$ we say that for any $m > 0$ and $b \in \mathbb{R}$, that the attribute is

- **location invariant** if

$$a(y_1 + b, y_2 + b, \dots, y_N + b) = a(y_1, y_2, \dots, y_N)$$

- **location equivariant** if

$$a(y_1 + b, y_2 + b, \dots, y_N + b) = a(y_1, y_2, \dots, y_N) + b$$

Example

The population average is location equivariant.

$$a(\mathcal{P}) = a(y_1, y_2, \dots, y_N) = \frac{1}{N} \sum_{i=1}^N y_i$$

$$\begin{aligned} a(y_1 + b, y_2 + b, \dots, y_N + b) &= \frac{1}{N} \sum_{i=1}^N (y_i + b) \\ &= \frac{1}{N} \sum_{i=1}^N y_i + \frac{Nb}{N} = a(\mathcal{P}) + b \end{aligned}$$

Scale Invariance and Equivariance

For an attribute $a(\mathcal{P}) = a(y_1, \dots, y_N)$ we say that for any $m > 0$ and $b \in \mathbb{R}$, that the attribute is

- **scale invariant** if

$$a(mxy_1, mxy_2, \dots, mxy_N) = a(y_1, y_2, \dots, y_N)$$

- **scale equivariant** if

$$a(mxy_1, mxy_2, \dots, mxy_N) = m \times a(y_1, y_2, \dots, y_N)$$

- **location-scale invariant** if it is both location invariant and scale invariant, i.e.

$$a(mxy_1 + b, mxy_2 + b, \dots, mxy_N + b) = a(y_1, y_2, \dots, y_N)$$

)

- **location-scale equivariant** if it is both location equivariant and scale equivariant, i.e.

$$a(mx_1 + b, mx_2 + b, \dots, mx_n + b) = m \times a(y_1, y_2, \dots, y_n) + b$$

Example

The population average is location-scale equivariant

$$\begin{aligned} a(my_1 + b, my_2 + b, \dots, my_n + b) &= \frac{1}{N} \sum_{i=1}^N (my_i + b) \\ &= \frac{m}{N} \sum_{i=1}^N y_i + \frac{N}{N} b \\ &= m a(y_1, y_2, \dots, y_n) + b \end{aligned}$$

Replication

- Another invariance/equivariance property of interest for population attributes is **replication invariance** and **replication equivariance**.

– If a population $\mathcal{P}$ is duplicated $k-1$ times (so that there are k copies of it), how does the attribute change on this new population denoted by $\mathcal{P}^k$?

$$\mathcal{P}^k = \{y_1, y_2, \dots, y_n, y_1, y_2, \dots, y_n, \dots, y_1, y_2, \dots, y_n\} = \{x_1, x_2, \dots, x_{nk}\}$$

- The attribute $a(\mathcal{P})$ is

- **replication invariant** whenever $a(\mathcal{P}^k) = a(\mathcal{P})$ and
- **replication equivariant** whenever $a(\mathcal{P}^k) = k a(\mathcal{P})$

Example

The population average is replication invariant.

$$a(P^K) = \frac{1}{NK} \sum_{j=1}^{NK} x_j = \frac{1}{NK} \sum_{i=1}^N K y_i = \frac{1}{N} \sum_{i=1}^N y_i = a(P)$$

Summary

- We examined invariance and equivariance when we change the units in some way, and when we replicate the population.
- We used the population average as an example.

Exercises

- From the Explicit Attributes exercises try problems 6, 8 and 10. For continued learning try problems 1, 2, 9, 11 and 20.

2.2.3 Influence, Sensitivity Curves, and Breakdown Points

Overview

- If we wish to evaluate a population attribute

$$a(\mathcal{P}) = a(y_1, \dots, y_N)$$

we could consider the effect of adding or removing a single unit from the population $\mathcal{P}$.

- To quantify this effect, we could look at the change in the attribute's value when the unit's variate value is added (sensitivity) or removed (influence).
- We also consider a special form of sensitivity called the breakdown point.

Influence

outlier
detection

* measures something
about the units

- If we remove variate y_u , (i.e. remove unit u) then the **influence** of that variate on the population attribute is quantified by

$$\Delta(a, u) = a(y_1, \dots, y_{u-1}, y_u, y_{u+1}, \dots, y_N) - a(y_1, \dots, y_{u-1}, y_{u+1}, \dots, y_N)$$

population with the unit u population without the unit u

- Ideally, no single unit's value should have greater influence than any other.
- If a unit has larger influence than the rest;
 - ① – it would require further investigation as it might be in error, or
 - ② – it might be the most interesting unit in the population.

Example

From the agricultural census, we can examine the number of farms per county in 1987.

The population average, $a(y_1, \dots, y_N) = \bar{y}$, is

```

directory <- "../..Data"
dirsep <- "/"
filename <- paste(directory, "agpop_data.csv", sep=dirsep)
agpop <- read.csv(filename, header=TRUE)

y = agpop$farms87
ybar = mean(y)
ybar

```

```
## [1] 678.2843 ←
```

and the average without unit u can be written as

$$a(y_1, \dots, y_{u-1}, y_{u+1}, \dots, y_N) = \frac{1}{N-1} \sum_{\substack{k \in P \\ k \neq u}} y_k = \frac{N\bar{y} - y_u}{N-1}$$

and $\Delta(a, u)$, the influence for a given u , is:

$$\Delta(a, u) = \bar{y} - \frac{N\bar{y} - y_u}{N-1} = \frac{y_u - \bar{y}}{N-1}$$

We can use R to calculate this quantity in many ways.

- Using a loop and calculating the Δ :

```

delta = rep(0, length(y)) ←
for (i in 1:length(y)) {
  ## y[-i] removes the ith element from a vector
  delta[i] = ybar - mean(y[-i])
}

```

- Creating a matrix of populations of size $N - 1$ and using the `apply` function:

```

N = length(y)
popN_1 = matrix(rev(y), nrow=N-1, ncol=N)
attN_1 = apply(popN_1, 2, mean)
delta = ybar - attN_1 ↑

```

- Using a loop and the simplified expression:

```

delta = rep(0, length(y))
for (i in 1:length(y)) {
  delta[i] = (y[i]-ybar)/(N-1)
}

```

- Summing two numeric vectors:

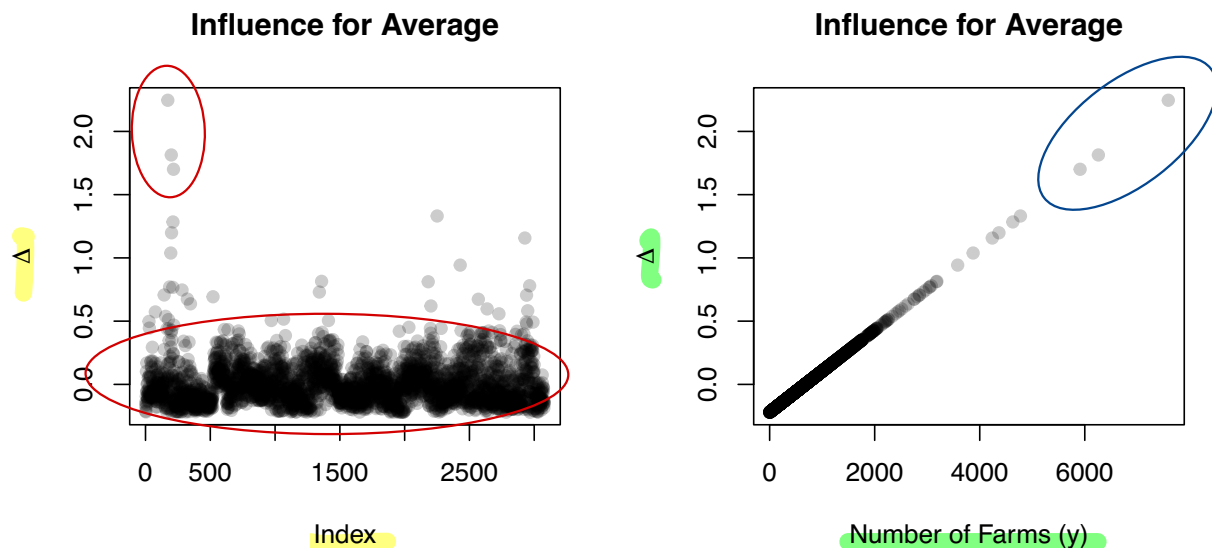
– Note: when a vector $\mathbf{v}$ of dimension N is added to a scalar c , R converts the scalar into $c\mathbf{1}_N$

✱ $\text{delta} = (y - \bar{y}) / (N - 1)$

To summarize the influence for every unit u we might plot the influence (Δ values) by observation number or by y .

```
y = agpop$farms87
N = length(y)
delta = sum(y)/N - (sum(y)-y)/(N-1)

par(mfrow=c(1,2))
plot(delta, main="Influence for Average", pch=19,
     col=adjustcolor("black", alpha = 0.2),
     xlab = "Index",
     ylab = bquote(Delta))
plot(y, delta, main="Influence for Average", pch=19,
     col=adjustcolor("black", alpha = 0.2),
     xlab='Number of Farms (y)',
     ylab = bquote(Delta))
```



- There is, at least, one (if not a few) counties whose farm numbers seem to be more influential on the population average compared to other counties.
- These counties are, as expected (why?), those with a larger number of farms.

✱ – it turns out that the three highest influence points are observation numbers 172 (Fresno), 199 (San Diego), and 216 (Tulare) with 7590, 6259, and 5911 farms, respectively. All these counties are in California.

– These are the three most extreme values in the right panel.

Sensitivity Curve

* measures something about the attribute

- We can also examine the effect on an attribute when we add a variate. To examine this effect,
 - suppose we have a population of size $N - 1$ and
 - add a variate with the value y .
 - Then our new population with N elements is $\{y_1, \dots, y_{N-1}, y\}$.
- We define the **sensitivity curve** of an attribute as

$$sc(y, a) = N [a(y_1, \dots, y_{N-1}, y) - a(y_1, \dots, y_{N-1})]$$

$\underbrace{\quad}_{P^*} \qquad \underbrace{\quad}_P$

- We can then plot the **sensitivity curve** as a function of the new variate value y .
 - the sensitivity curve gives a scaled measure of the effect that a single variate value y has on the value of a population attribute $a(P)$.
 $N \approx 1 \div 1/N$ } proportion of new data
- We can explore the sensitivity curve for any attribute. These can be determined **mathematically** in general, but can also be determined **computationally** for any particular population and any particular attribute.
- The following is a general-purpose sensitivity curve function in R which accommodates any population and any attribute:

```
sc = function(y.pop, y, attr, ...) {
  N <- length(y.pop) + 1
  sapply(y, function(y.new) { N*(attr(c(y.new, y.pop), ...) - attr(y.pop, ...)) } )
}
```

Example: Arithmetic Mean

$$a(y_1, \dots, y_N) = \frac{1}{N} \sum_{i=1}^N y_i = \bar{y}$$

- Derive the sensitivity curve.

$$P = \{y_1, \dots, y_{N-1}\}$$

$$P^* = \{y_1, \dots, y_{N-1}, y\}$$

$$a(P) = \frac{1}{N-1} \sum_{i=1}^{N-1} y_i = \bar{y}_{N-1}$$

$$a(P^*) = \frac{1}{N} \left[\sum_{i=1}^{N-1} y_i + y \right] = \frac{(N-1)\bar{y}_{N-1} + y}{N}$$

$$\begin{aligned}
 \therefore SC(a,u) &= N [a(P^*) - a(P)] \\
 &= N \left[\frac{(N-1)\bar{y}_{N-1} + y}{N} - \bar{y}_{N-1} \right] \\
 &= y - \bar{y}_{N-1}
 \end{aligned}$$

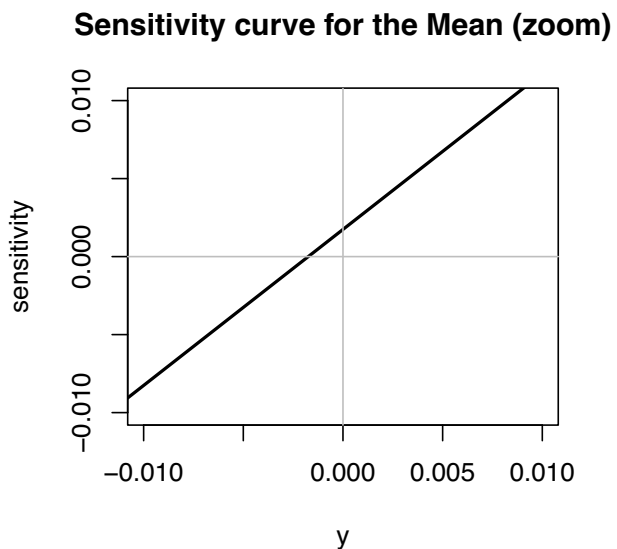
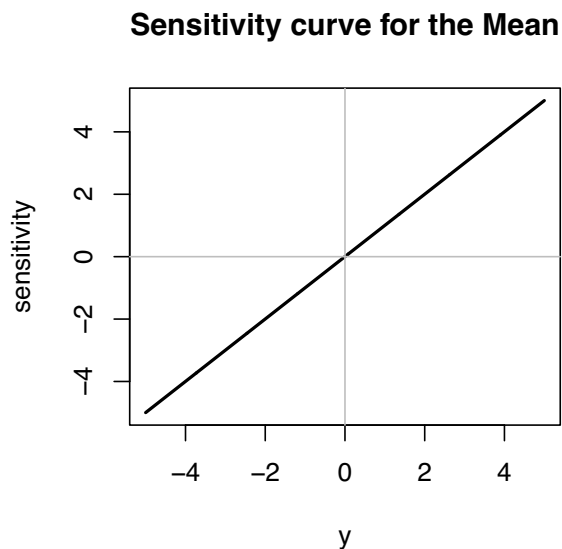
- We found the sensitivity curve to be $SC(y) = y - \bar{y}_{N-1}$
- The sensitivity curve for the arithmetic average is

```

set.seed(341)
P <- rnorm(1000)
y <- seq(-5,5, length.out=1000)
par(mfrow=c(1,2))
plot(y, sc(P, y, mean), type="l", lwd = 2,
     main="Sensitivity curve for the Mean",
     ylab="sensitivity")
abline(h=0, v=0, col="grey")

plot(y, sc(P, y, mean), type="l", lwd = 2,
     main="Sensitivity curve for the Mean (zoom)",
     ylab="sensitivity",
     xlim=c(-.01,.01),ylim=c(-.01,.01))
abline(h=0, v=0, col="grey")

```



Note

- The sensitivity curve here gets higher (or lower) without bound as $y \rightarrow \infty$ (or as $y \rightarrow -\infty$).
- A single observation can change the average by a huge (even infinite) amount.
- Averages may not be the best choice for a population attribute representing the location of a population – particularly if extreme values exist in the population.

Example: Maximum

$$a(y_1, \dots, y_N) = \max \{y_1, \dots, y_N\} = y_{(N)}$$

- Derive the sensitivity curve.

$$P = \{y_1, \dots, y_{N-1}\}$$

$$P^* = \{y_1, \dots, y_{N-1}, y\}$$

$$\begin{aligned} a(P) &= \max \{y_1, \dots, y_{N-1}\} \\ &= y_{(N-1)} \end{aligned}$$

$$\begin{aligned} a(P^*) &= \max \{y_1, \dots, y_{N-1}, y\} \\ &= \begin{cases} y_{(N-1)} & \text{if } y < \max \{y_1, \dots, y_{N-1}\} \\ y & \text{if } y \geq \max \{y_1, \dots, y_{N-1}\} \end{cases} \end{aligned}$$

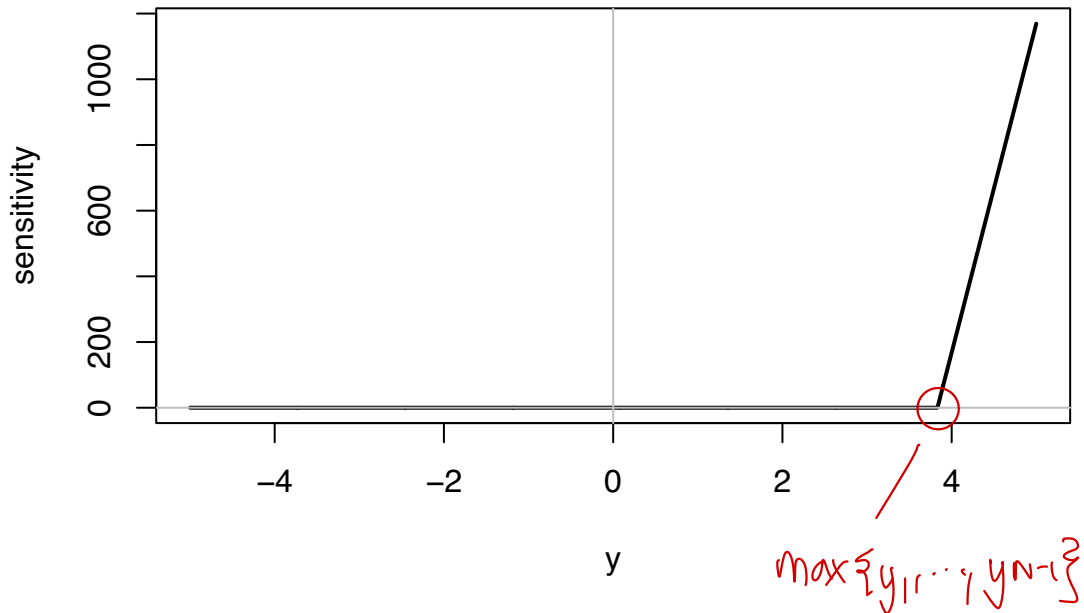
$$\therefore sc(y, a) = N [a(P^*) - a(P)]$$

$$= \begin{cases} 0 & \text{if } y < \max \{y_1, \dots, y_{N-1}\} \\ N[y - y_{(N-1)}] & \text{if } y \geq \max \{y_1, \dots, y_{N-1}\} \end{cases}$$

- The sensitivity curve for the maximum is

```
plot(y, sc(P, y, max), type="l", lwd = 2,
     main="Sensitivity curve for the Maximum",
     ylab="sensitivity")
abline(h=0, v=0, col="grey")
```

Sensitivity curve for the Maximum



How do you interpret this curve?

The fact that it is unbounded for large y means that the maximum is very sensitive to large outliers.

Example: 2nd Order Statistic

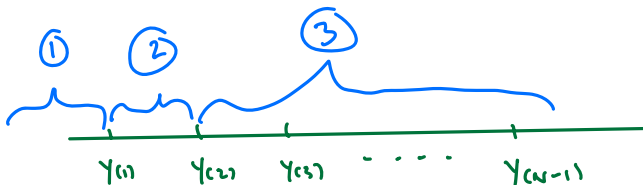
$$\underline{a(y_1, \dots, y_N) = y_{(2)}}$$

- Derive the sensitivity curve.

$$P = \{y_1, y_2, \dots, y_{N-1}\}$$



$$a(P) = y_{(2)}$$



$$P^* = \{y_1, y_2, \dots, y_{N-1}, y\}$$



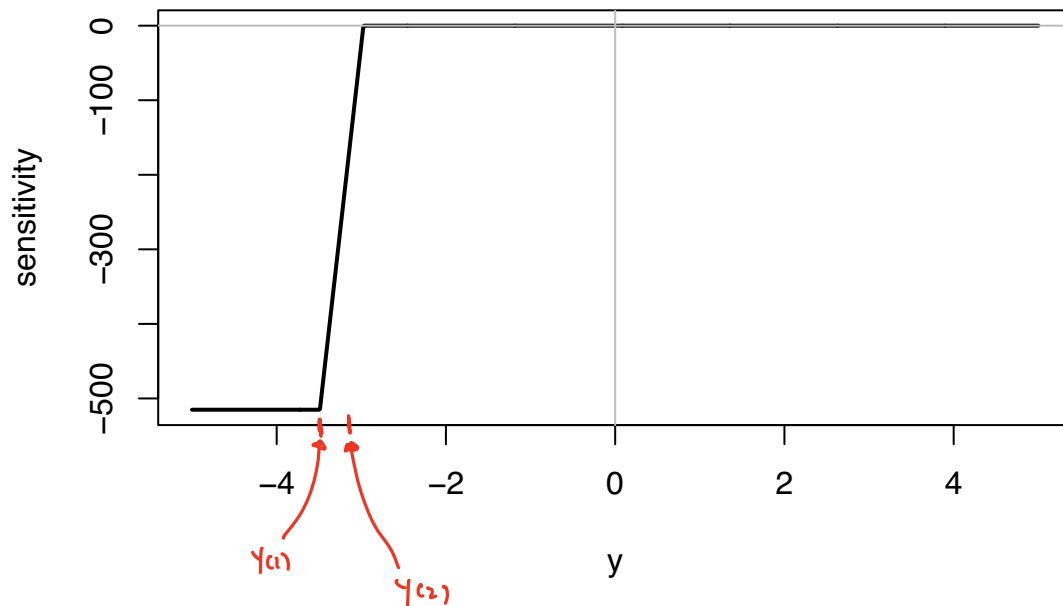
$$a(P^*) = \begin{cases} y_{(1)} & \text{if } y < y_{(1)} \quad \textcircled{1} \\ y & \text{if } y_{(1)} \leq y \leq y_{(2)} \quad \textcircled{2} \\ y_{(2)} & \text{if } y > y_{(2)} \quad \textcircled{3} \end{cases}$$

$$sc(y, a) = N[a(P^*) - a(P)] = \begin{cases} N(y_{(1)} - y_{(2)}) & \text{if } y < y_{(1)} \quad \textcircled{1} \\ N(y - y_{(2)}) & \text{if } y_{(1)} \leq y \leq y_{(2)} \quad \textcircled{2} \\ 0 & \text{if } y > y_{(2)} \quad \textcircled{3} \end{cases}$$

- 2nd Order Statistic - Sensitivity Curve

```
order.stat <- function(pop, k=1) { sort(pop)[k] }
plot(y, sc(P, y, order.stat, k=2), type="l", lwd = 2,
     main="Sensitivity curve for the 2nd smallest value",
     ylab="sensitivity")
abline(h=0, v=0, col="grey")
```

Sensitivity curve for the 2nd smallest value

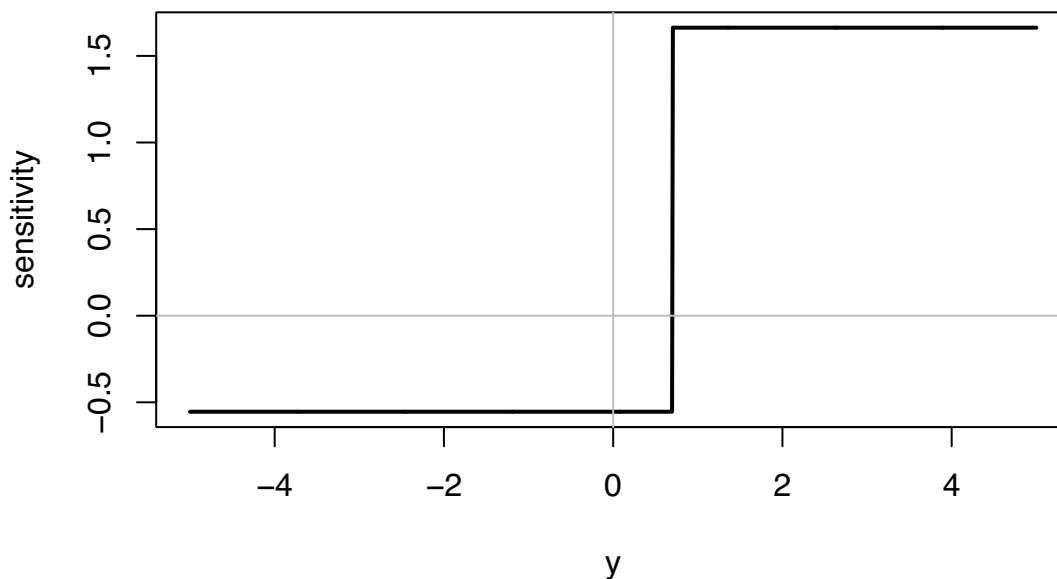


Example: 3rd Quartile

- Third Quartile - Sensitivity Curve

```
plot(y, sc(P, y, quantile, p=0.75), type="l", lwd = 2,
     main="Sensitivity curve for the Third Quartile",
     ylab="sensitivity")
abline(h=0, v=0, col="grey")
```

Sensitivity curve for the Third Quartile



Example: Median

Suppose the size of the original population (excluding the y value) is an even number $N - 1 = 2m$. Then the median is

$$a_{N-1}(y_1, \dots, y_{N-1}) = \frac{1}{2} (y_{(m)} + y_{(m+1)}).$$

Now, adding the new value y , the median $a_N(y_1, \dots, y_N)$ is calculated based on $N = 2m + 1$ values. The ordered values are

$$y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(N)}$$

and the sensitivity curve for the median is (**show this**)

$$SC(y) = \begin{cases} -\frac{N}{2} (y_{(m+1)} - y_{(m)}) & \text{if } y < y_{(m)} \\ \frac{N}{2} (2y - y_{(m+1)} - y_{(m)}) & \text{if } \underline{y_{(m)}} \leq y \leq \underline{y_{(m+1)}} \\ \frac{N}{2} (y_{(m+1)} - y_{(m)}) & \text{if } y > y_{(m+1)} \end{cases}$$

which looks like a

- negative constant when $y < y_{(m)}$,
- a positive constant at $y_{(m+1)} < y$,
- and a simple straight line with positive slope when y is between $y_{(m)}$ and $y_{(m+1)}$.

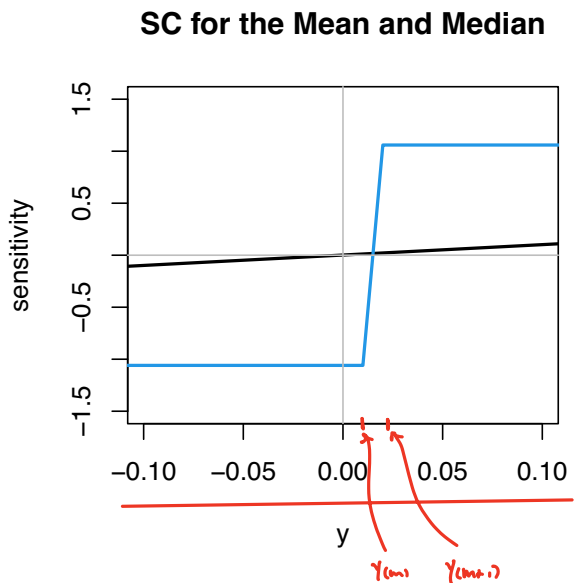
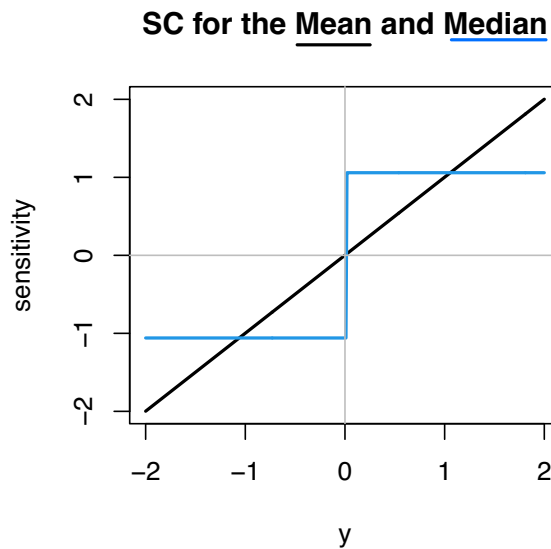
Exercise: Try doing this for odd $N-1$

As with the arithmetic average, we can draw the sensitivity curve now for the median for any particular sample.

Example: Comparing the Median & Mean SC Plot

```
par(mfrow=c(1,2))
y = seq(-2,2,.01)
plot(y, sc(P, y, mean), type="l", lwd = 2,
     main="SC for the Mean and Median",
     ylab="sensitivity")
lines(y, sc(P, y, median), lwd=2, col=4)
abline(h=0, v=0, col="grey")

plot(y, sc(P, y, mean), type="l", lwd = 2,
     main="SC for the Mean and Median",
     ylab="sensitivity",xlim=c(-.1,.1),ylim=c(-1.5,1.5))
lines(y, sc(P, y, median), lwd=2, col=4)
abline(h=0, v=0, col="grey")
```



- Unlike the arithmetic mean, the sensitivity curve for the median is at least bounded.
 - A single observation cannot change the median by very much (“robust” to outliers).
 - This makes the median an interesting population location attribute.

Breakdown Points

- Another measure of robustness that exists is called the breakdown point.
 - It gives an assessment of just how large a proportion of the data must be contaminated before the statistic breaks down (and becomes useless).

Suppose we have the following population

```
x = c(22,5,3,2,21)
x
```

```
## [1] 22 5 3 2 21
```

We can calculate the mean and median:

```
c(mean(x), median(x))
```

```
## [1] 10.6 5.0
```

Then if we change the first value in the population to infinity the mean and median become:

```
y = x
y[1] = Inf
c(mean(y), median(y))
```

```
## [1] Inf 5
```

The difference between the two mean and median calculations (called **error**) is:

```
c(mean(y), median(y)) - c(mean(x), median(x))
```

```
## [1] Inf 0
```

* The breakdown point of a statistic is the *smallest possible fraction* of the observations that can be changed to something very extreme (i.e., plus or minus infinity) to make the error large (infinite)

- e.g. the break-point for
 - the average is $1/N$ (or asymptotically zero), and
 - the median is $1/2$ (i.e., that is half of the data has to go to infinity before the median breaks down).
- Attributes with high breakdown points are called **resistant** or **robust**.

Summary

- We examined *influence*, the effect on an attribute when we remove a unit from the population.
 - We typically use this to find influential units in a population.
- We also examined *sensitivity*, the effect on an attribute when we add a unit to the population.
 - We typically use this to investigate the robustness of an attribute to extreme variate values.
- We looked at another measure of robustness called the breakdown point.

Exercises

From the Explicit Attribute exercises try problems 3 and 4. For continued learning try problems 5, 12, 16, 18 and 23.

2.2.4 Graphical Attributes

Overview

- We investigate graphical population attributes.

Population attributes can also be entirely graphical as in

- histograms of y_u values
 - bar plots of y_u values
 - box plots of y_u values
 - scatter-plots of pairs (x_u, y_u)
 - scatter-plots of quantiles and ranks of y_u .
- univariate graphical summaries*
- bivariate graphical summaries*

✦ Each of these plots summarizes the entire population, and so they're all attributes.

Histograms

- Consider the population $\mathcal{P} = \{y_1, y_2, \dots, y_N\}$.
 - Partition the range of the population into k non-overlapping intervals, called **bins**, $I_j = [a_{j-1}, a_j)$, $j = 1, 2, \dots, k$ and then calculate the number or proportion of observations in the j^{th} bin for $j = 1, \dots, k$.
- ✦ Histograms help determine how the values are concentrated.

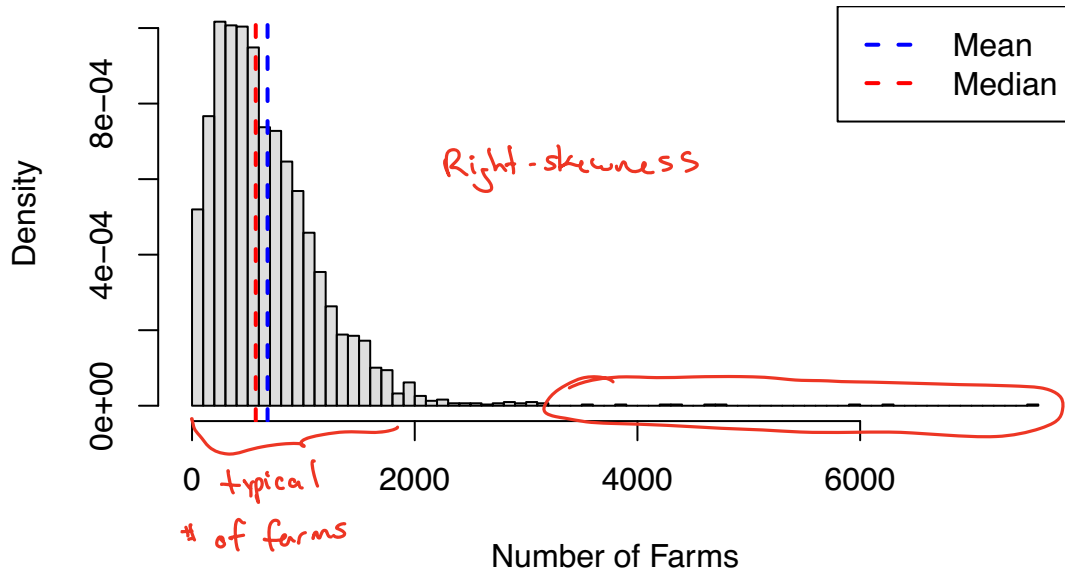
```
directory <- "../Data"
dirsep <- "/"
filename <- paste(directory, "agpop_data.csv", sep=dirsep)
agpop <- read.csv(filename, header=TRUE)
agpop[agpop == -99] = NA
```

```
hist(agpop$farms87, col=adjustcolor("grey", alpha = 0.5),
     main="Number of Farms per County in 1987",
     xlab="Number of Farms",
     breaks=100, prob=TRUE)
```

```
abline(v=c(mean(agpop$farms87), median(agpop$farms87)),
       col=c("blue", "red"), lwd=2, lty=2)
```

```
legend("topright", c("Mean", "Median"), lwd=2, lty=2, col=c("blue", "red"))
```

Number of Farms per County in 1987



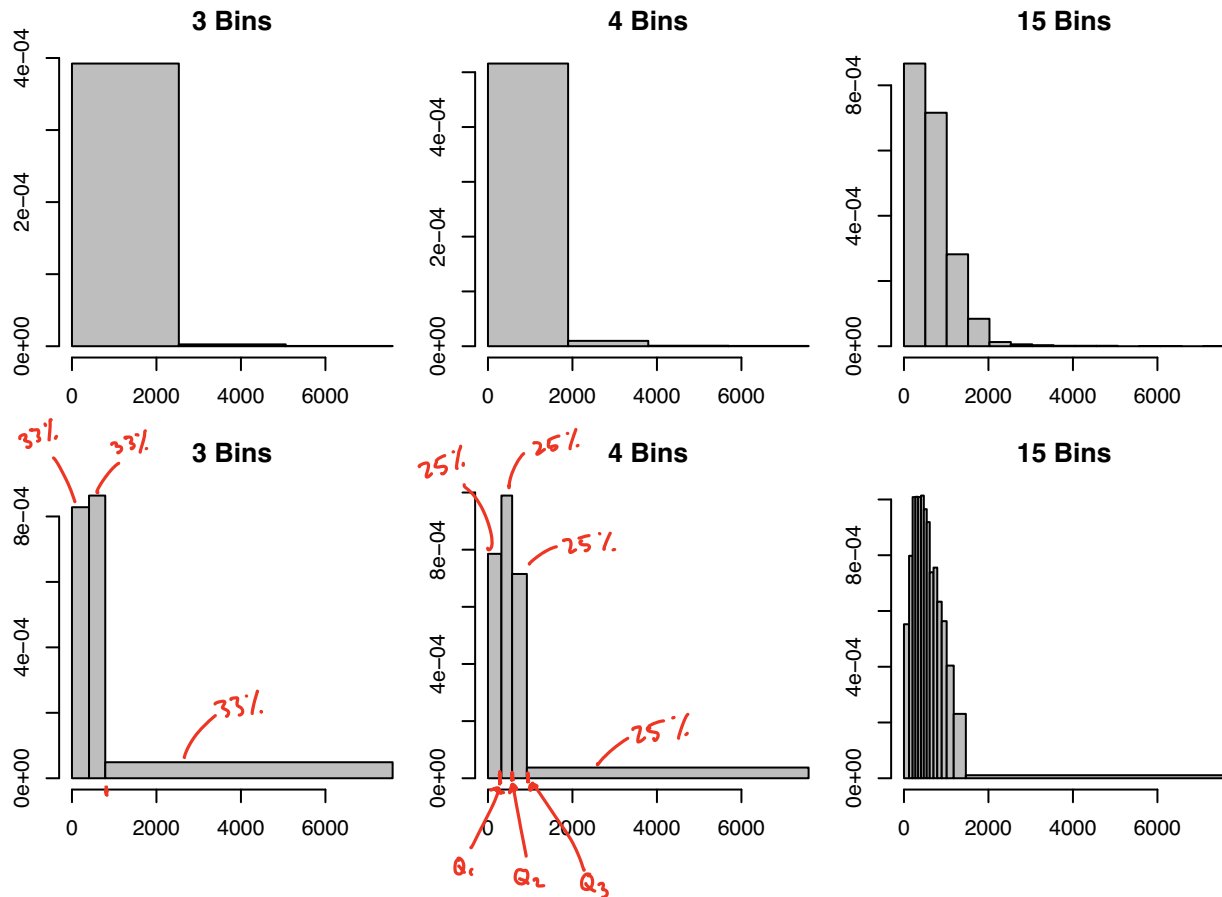
- We can define bins two ways:

→ bins of equal size, or (this is most common)
 → bins with equal number of elements but varying size. ← "Equal Area" histogram

- Below are some examples of histograms with equal-sized bins (top row) and bins of varying sizes (bottom row)

```
x = agpop$farms87
par(mfrow=c(2,3), mar=2.5*c(1,1,1,0.1))
rx = range(x)
hist(x, breaks=seq(rx[1], rx[2], length.out=4), prob=TRUE, main="3 Bins", col = "grey")
hist(x, breaks=seq(rx[1], rx[2], length.out=5), prob=TRUE, main="4 Bins", col = "grey")
hist(x, breaks=seq(rx[1], rx[2], length.out=16), prob=TRUE, main="15 Bins", col = "grey")

hist(x, breaks=quantile(x, p=seq(0, 1, length.out=4)), prob=TRUE, main="3 Bins", col = "grey")
hist(x, breaks=quantile(x, p=seq(0, 1, length.out=5)), prob=TRUE, main="4 Bins", col = "grey")
hist(x, breaks=quantile(x, p=seq(0, 1, length.out=16)), prob=TRUE, main="15 Bins", col = "grey")
```



- Notice also how varying the bin size changes the coarseness of the histogram.
- For the histograms in the bottom row, the areas of all rectangles in each panel are the same.
- The bins with equal numbers of elements but varying size can help identify asymmetry in the population.

Rules for the Number of Bins (when bin sizes are the same)

- Sturges rule:

the number of bins should be $= \lceil \log_2(N) + 1 \rceil$ ← default in R

- Freedman–Diaconis rule:

$$\text{Bin size} = 2 \frac{\text{IQR}(x)}{N^{1/3}}$$

- Scott's rule:

$$\text{Bin size} = 3.5 \frac{\sigma}{N^{1/3}}$$

- Histograms using different rules for bin size selection:

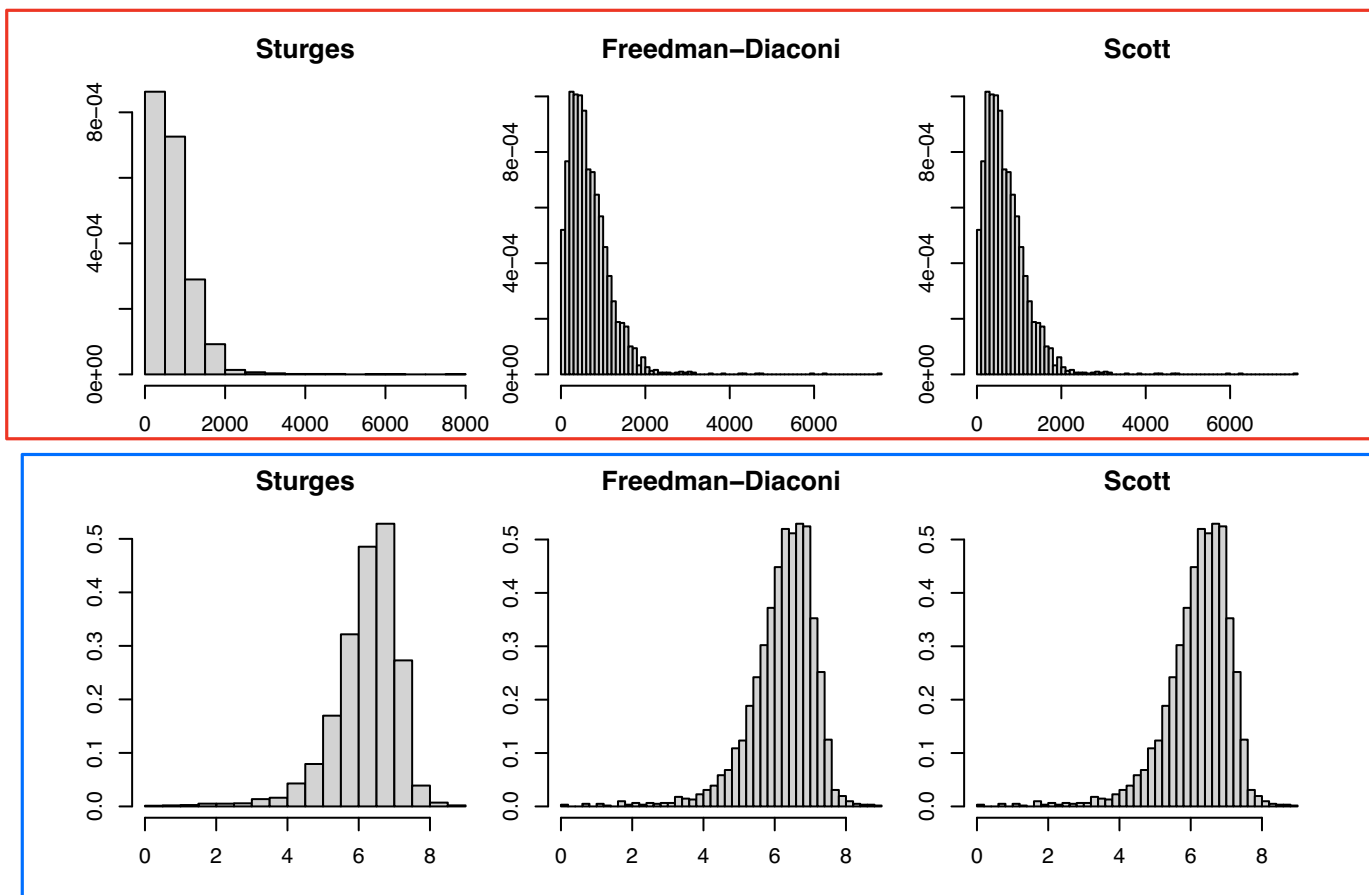
- the first row is Number of farms and
- the second row is $\log(\text{Number of farms}+1)$.

```

par(mfrow=c(2,3), mar=2.5*c(1,1,1,0.1))
x = agpop$farms87
hist(x, prob=TRUE, xlab="", main="Sturges")
hist(x, breaks="FD", prob=TRUE, xlab="", main="Freedman-Diaconi")
hist(x, breaks="scott", prob=TRUE, xlab="", main="Scott")

x = log(agpop$farms87 +1)
hist(x, prob=TRUE, xlab="", main="Sturges")
hist(x, breaks="FD", prob=TRUE, xlab="", main="Freedman-Diaconi")
hist(x, breaks="scott", prob=TRUE, xlab="", main="Scott")

```



- **Aside:** Which scale would you prefer to work with? The original scale or the transformed scale?

Raw Data: advantage is that it's more easily interpreted

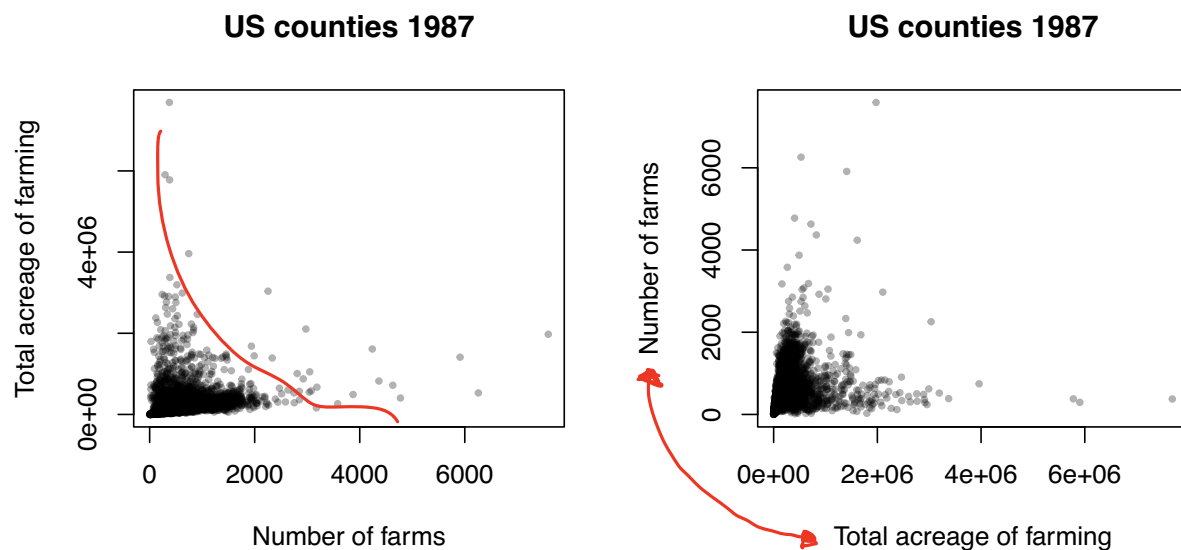
Transformed Data: advantage is that symmetric data can be easier to work with, statistically speaking

Scatter-plots

- A scatter-plot is a plot of the points (x_u, y_u) for all units in the population.
 - It is used to see whether two variates x and y are related in some way
- A scatter-plot of the number of farms and total acreage of farming in 1987 by US county is below.

```
par(mfrow=c(1,2))
plot(agpop$farms87, agpop$acres87, pch = 19, cex=0.5,
     col=adjustcolor("black", alpha = 0.3),
     xlab = "Number of farms", ylab = "Total acreage of farming",
     main = "US counties 1987")

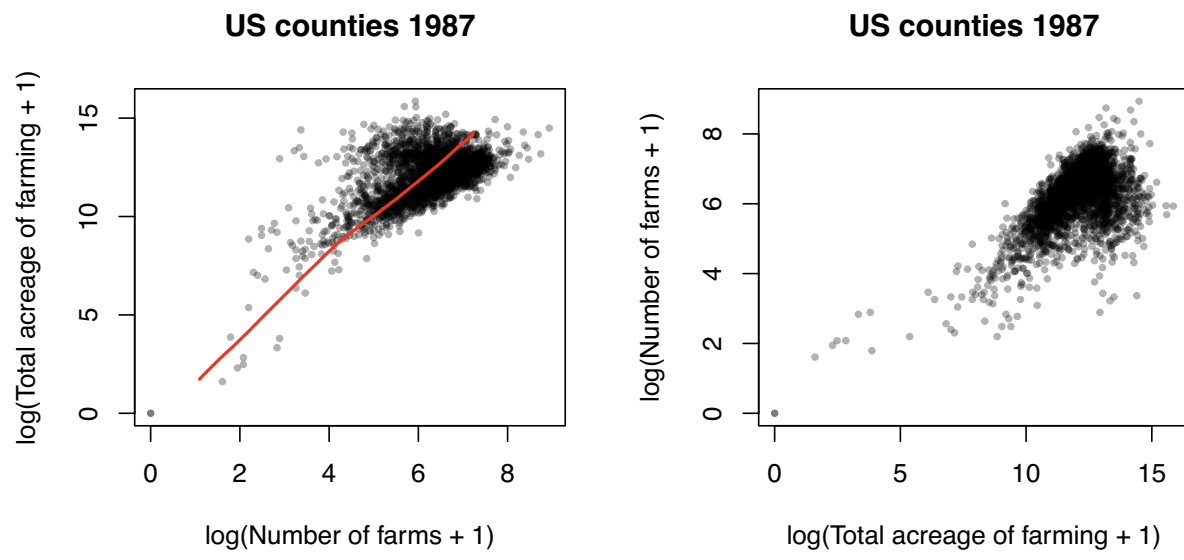
plot(agpop$acres87, agpop$farms87, pch = 19, cex=0.5,
     col=adjustcolor("black", alpha = 0.3),
     ylab = "Number of farms", xlab = "Total acreage of farming",
     main = "US counties 1987")
```



- Sometimes, the scatter-plot of a transformed version of the data provides more insight. We will discuss this later.

```
par(mfrow=c(1,2))
plot(log(agpop$farms87+1), log(agpop$acres87+1), pch = 19, cex=0.5,
     col=adjustcolor("black", alpha = 0.3),
     xlab = "log(Number of farms + 1)", ylab = "log(Total acreage of farming + 1)",
     main = "US counties 1987")

plot(log(agpop$acres87+1), log(agpop$farms87+1), pch = 19, cex=0.5,
     col=adjustcolor("black", alpha = 0.3),
     ylab = "log(Number of farms + 1)", xlab = "log(Total acreage of farming + 1)",
     main = "US counties 1987")
```



Example: Fire Emblem Heroes



Fire Emblem Heroes is a free-to-play tactical role-playing game developed by Intelligent Systems and Nintendo for iOS and Android devices. This population has $N = 168$ characters with varying types and movements.

- Some variates are shown below but the game mechanics allow for the development of new variates.

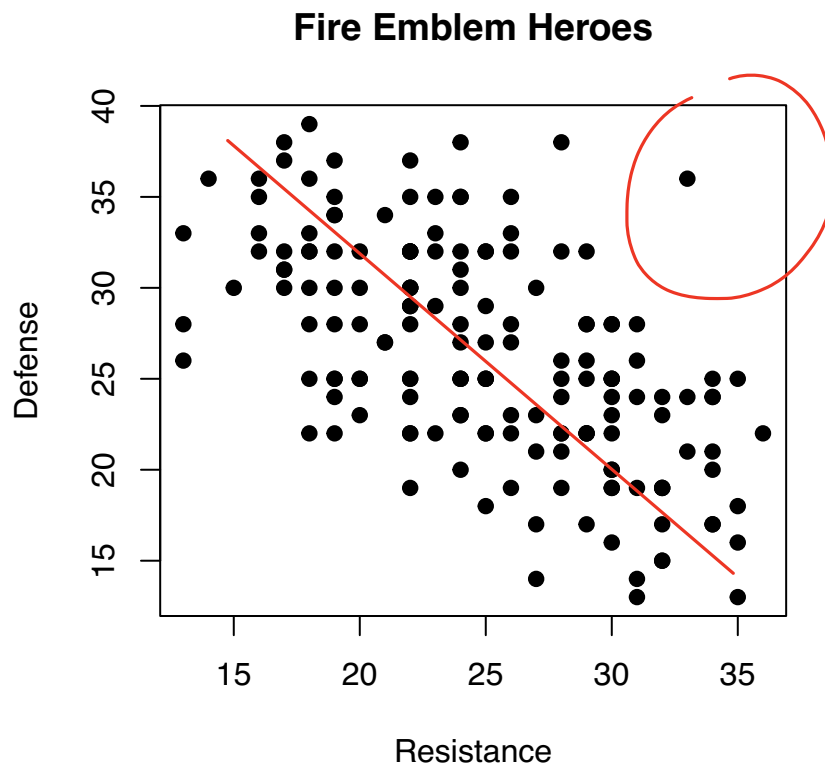
```
feh <- read.csv("../Data/feh.csv", header = TRUE)
head(feh)
```

##	Name	Type	Move	HP	ATK	SPD	DEF	RES	Total
## 1	Abel	Blue Lance	Cavalry	39	33	32	25	25	154
## 2	Alfonse	Red Sword	Infantry	43	35	25	32	22	157
## 3	Alm	Red Sword	Infantry	45	33	30	28	22	158
## 4	Amelia	Green Axe	Armored	47	34	34	35	23	173
## 5	Anna	Green Axe	Infantry	41	29	38	22	28	158
## 6	Arthur	Green Axe	Infantry	43	32	29	30	24	158

- Here we will consider the following two variates in particular:

- RES: Resistance, the ability to absorb magical attacks
- DEF: Defense, the ability to absorb physical attacks

```
plot(feh$RES, feh$DEF, main="Fire Emblem Heroes", pch = 19, cex=1,
     col=adjustcolor("black", alpha = 1), xlab="Resistance", ylab="Defense" )
```



- Problem: The variates are integer-valued making duplicate values difficult to identify.
- Solution: Try changing the shading and varying the size of bullets.

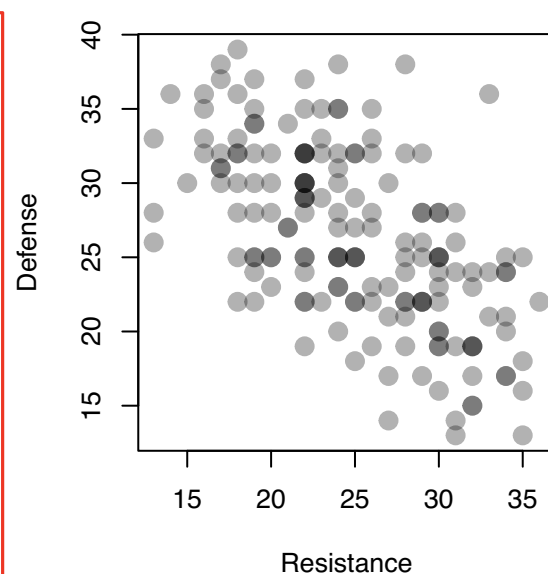
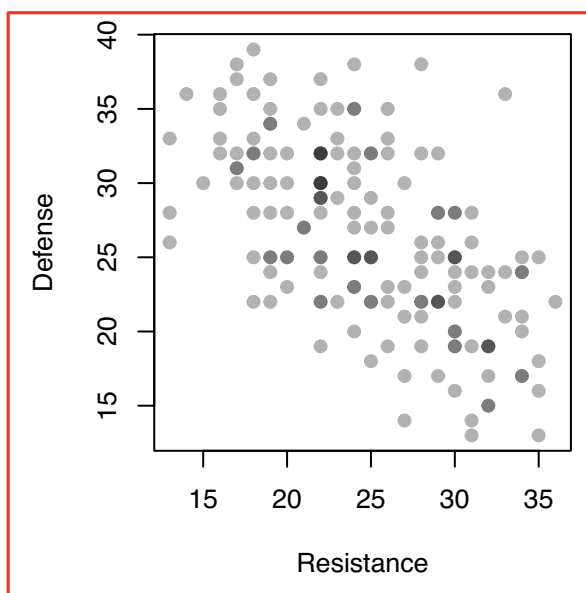
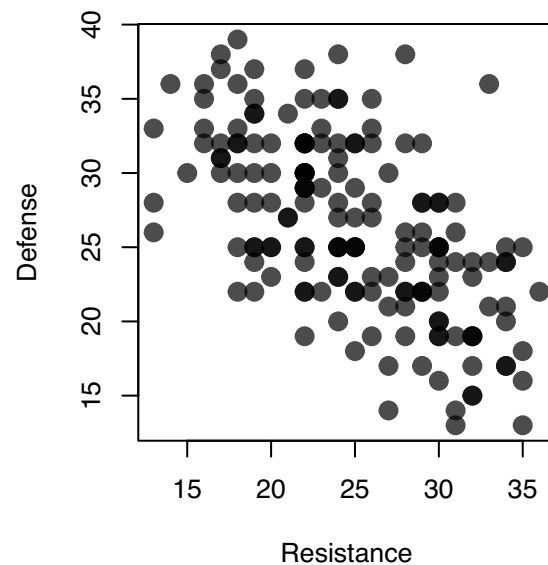
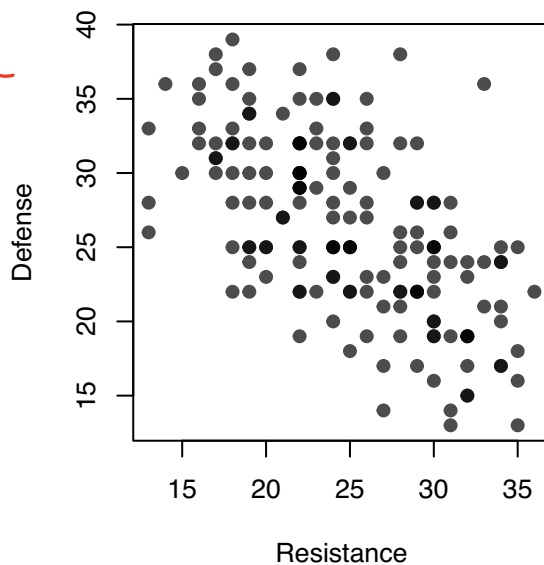
```
par(mfrow=c(2,2), mar=c(4,4,1,1))

cex.seq = rep( c(1,1.5), times=2)
shade.seq = rep( c(0.7,0.3), each=2)

for (i in 1:4) {
  plot(feh$RES, feh$DEF, main="", pch = 19,
       cex=cex.seq[i],
       col=adjustcolor("black", alpha = shade.seq[i]),
       xlab="Resistance", ylab="Defense" )
}
```

Changes Size

changes shading



- Note the use of par to display four figures in one plot

- `mfrow=c(2,2)` makes 2x2 panel of figures.
- `mar=c(4,4,1,1)` changes the spacing so that the plots are more compact.
- Try changing or removing the `mar` and/or `mfrow` calls to see what kind of control you have.
- Another way to deal with discreteness (multiple points at the same coordinate) is to add jitter to the population values.
 - Jitter separates duplicate points slightly (provided it makes sense to do so).

$$y_u^* = y_u + \text{noise}$$

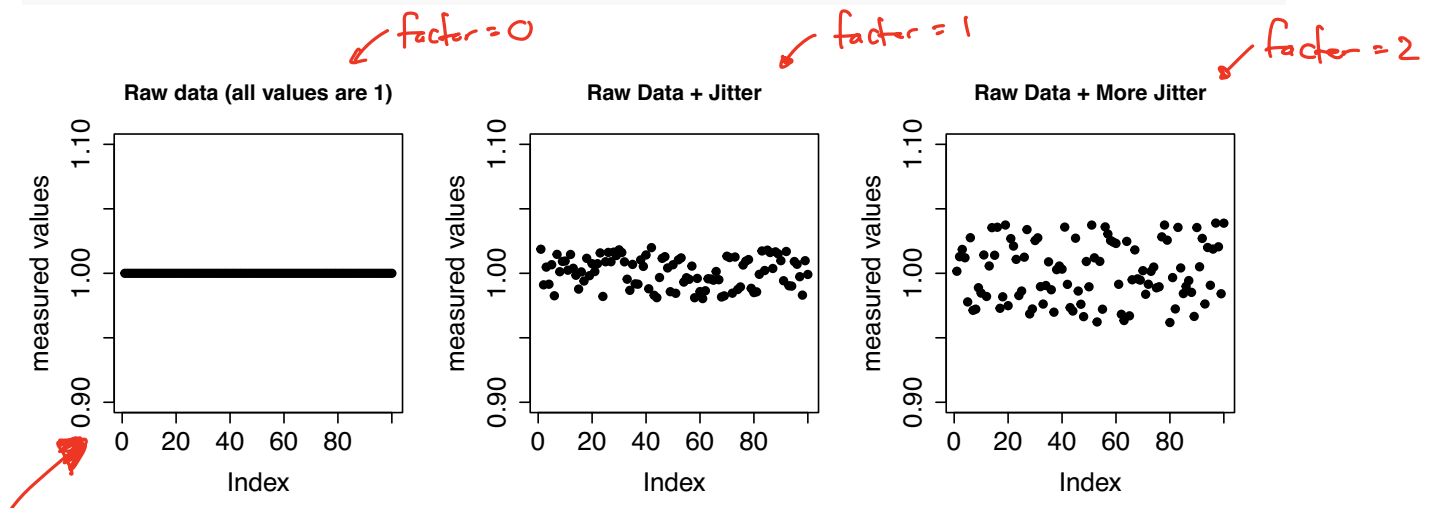
$N(0, \text{factor})$

- The amount of jitter can vary:

```
y.example <- rep(1,100)
par(mfrow=c(1,3)) #dividing the panel into 1 row and 3 columns for 3 plots

title.seq = c('Raw data (all values are 1)', 'Raw Data + Jitter', 'Raw Data + More Jitter')
fact.seq = c(0, 1, 2)

for (i in 1:3) {
  plot(jitter(y.example, factor= fact.seq[i]),
       main=title.seq[i],
       ylim=c(0.9,1.1), ylab='measured values',
       cex.lab=1.5, cex.axis=1.5, pch=19)
}
```

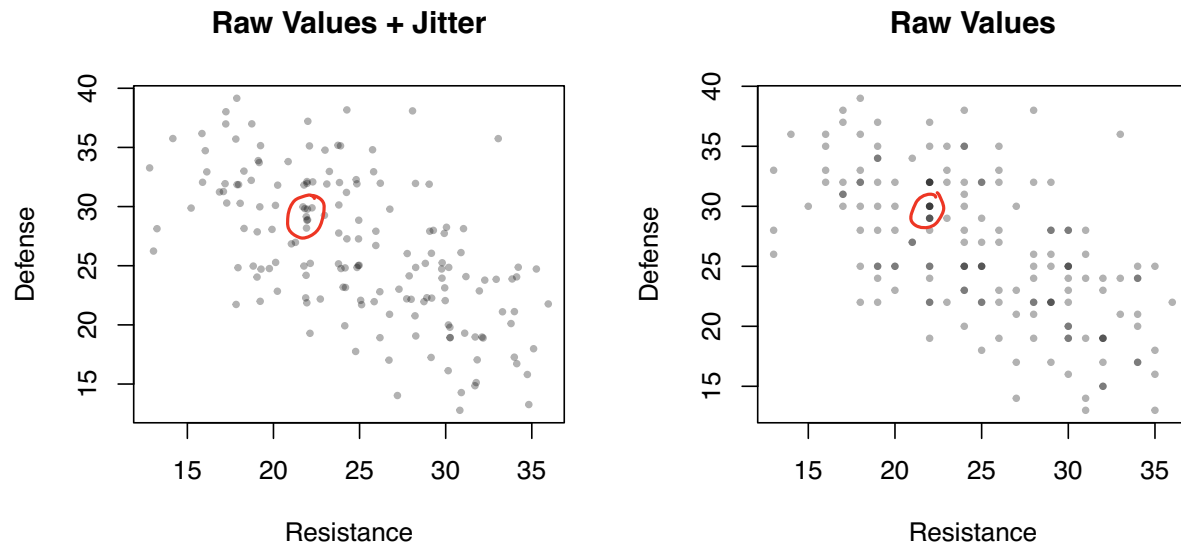


- Check the help documentation of the function `jitter` by typing `?jitter` or `help(jitter)` in R.
- The scatter-plot of Fire Emblem Heroes data with jitter is:

```
par(mfrow=c(1,2))

plot(jitter(feh$RES, factor=1.5), jitter(feh$DEF, factor=1.5),
     main="Raw Values + Jitter", pch = 19, cex=0.5,
     col=adjustcolor("black", alpha = 0.3),
     xlab="Resistance", ylab="Defense", type="p" )
```

```
plot(feh$RES, feh$DEF,
     main= "Raw Values",
     pch = 19, cex=0.5,
     col=adjustcolor("black", alpha = 0.3 ),
     xlab="Resistance", ylab="Defense" )
```



what's better, left or right?

- Advantage of left: more clarity in terms of how much data there is
- Advantage of Right: more clarity in terms of the data values themselves

* Jittering is acceptable when the goal is to improve the interpretability of a plot, but jittered data should not be analyzed numerically.

Summary

- We reviewed histograms and examined various rules for the number of bins in a histogram.
- We reviewed scatter-plots and examined different visualization strategies for making them more interpretable.

Exercises

From the Explicit Attribute exercises try problems 15 and 21.

2.2.5 Power Transformations

Overview

- We examine how the family of non-linear power transformations can be helpfully used to re-express a population of variate values so that on the transformed scale location/scale attributes are easier to define, to understand, or simply to determine.
- We explore two rules for picking an appropriate power transformation.

- For any variate y , it is sometimes helpful to re-express the values in a non-linear way via a transformation $T(y)$ so that on the transformed scale location/scale attributes are easier to define, to understand, or simply to determine.
- A commonly used transformation when $y > 0$ is the family of power transformations which is indexed by a power α . The general form is

$$T_{\alpha}(y) = \begin{cases} y^{\alpha} & \text{for } \alpha > 0 \\ \log(y) & \text{for } \alpha = 0 \end{cases}$$

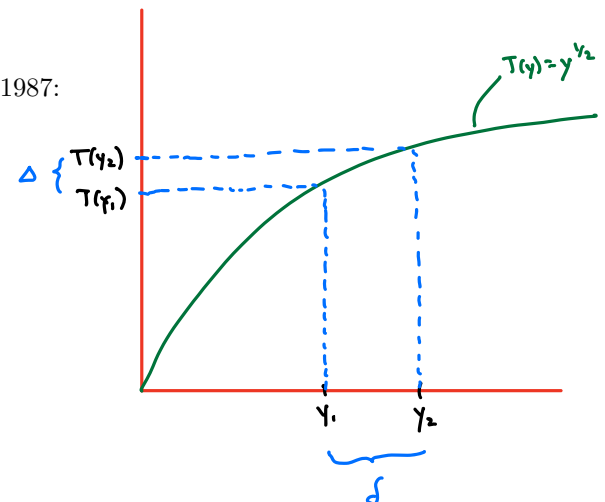
- These transformations are monotonic, in the sense that

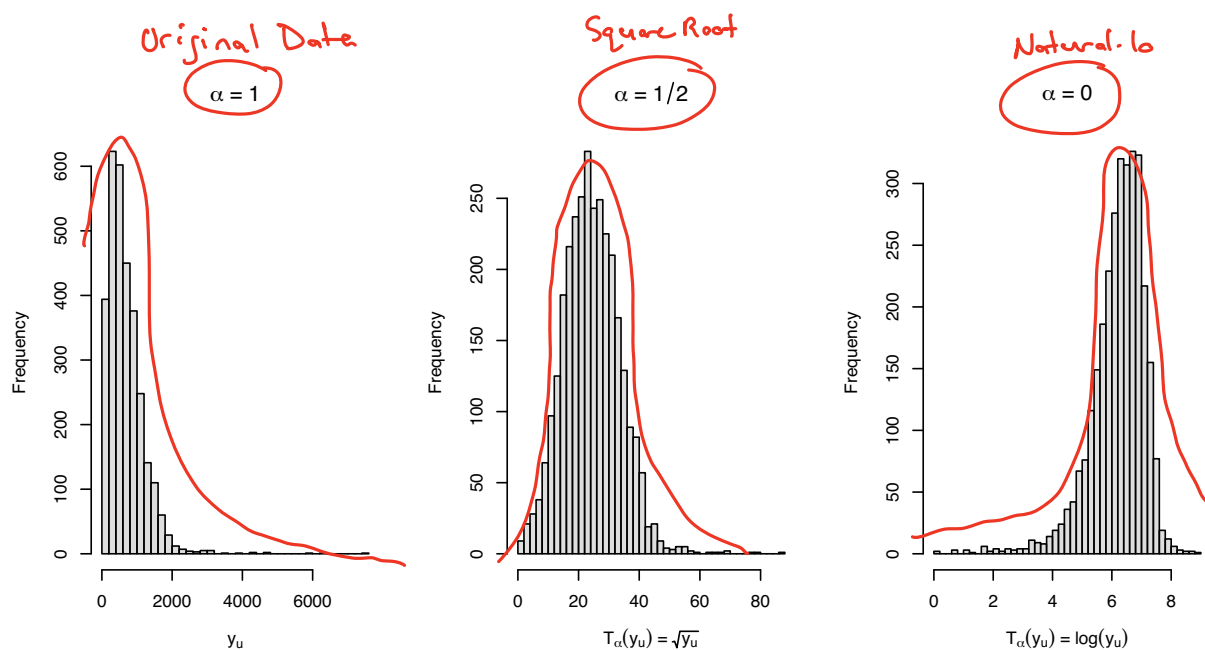
$$y_u < y_v \iff T_{\alpha}(y_u) < T_{\alpha}(y_v)$$

That is, they preserve the order of the variate values associated with the units u and v .

- What does change, often dramatically, is the relative positions of the variate values.

- The effect of the power transformation on number of farms in 1987:





* What is the effect of varying the power transformation?

• Different values of α change the "spacing" between observations

• Changing the spacing impacts the symmetry of the histogram

• **Note:** the most common purpose of a transformation is to change the shape of the histogram so that it is more symmetric.

– We mentioned that if $y > 0$, the family of **power transformations** indexed by a power α is defined as

$$T_{\alpha}(y) = \begin{cases} y^{\alpha} & \alpha \neq 0 \\ \log(y) & \alpha = 0 \end{cases}$$

• An alternative mathematical form is

$$T_{\alpha}(y) = \frac{y^{\alpha} - 1}{\alpha} \quad \forall \alpha$$

← "Box-Cox Transformation"

Note that the following limit gives rise to the $\alpha = 0$ case above:

$$\lim_{\alpha \rightarrow 0} T_{\alpha}(y) = \log(y)$$

• Yet another power transformation specification (with minimal potential for calculation errors) is the following:

$$T_{\alpha}(y) = \begin{cases} y^{\alpha} & \alpha > 0 \\ \log(y) & \alpha = 0 \\ -(y^{\alpha}) & \alpha < 0 \end{cases}$$

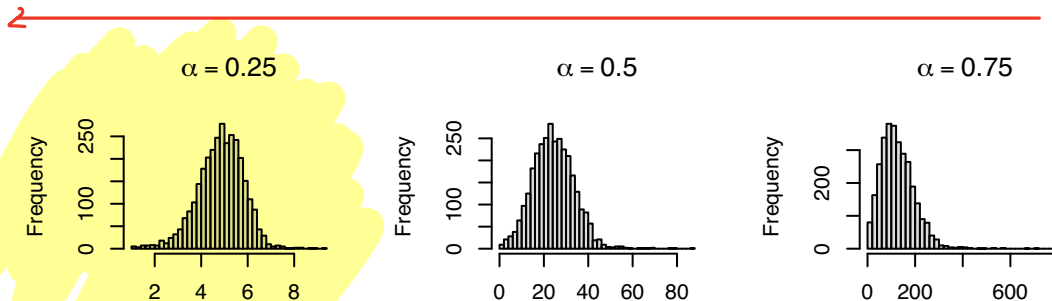
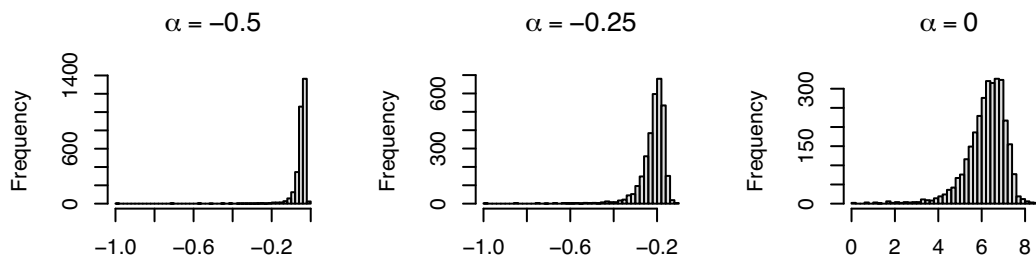
• A general-purpose R function to perform the power transformation is

```
powerfun <- function(y, alpha) {
  if(sum(y <= 0) > 0) stop("y must be positive")
  if (alpha == 0)
    log(y)
  else if (alpha > 0) {
    y^alpha
  } else -y^alpha
}
```

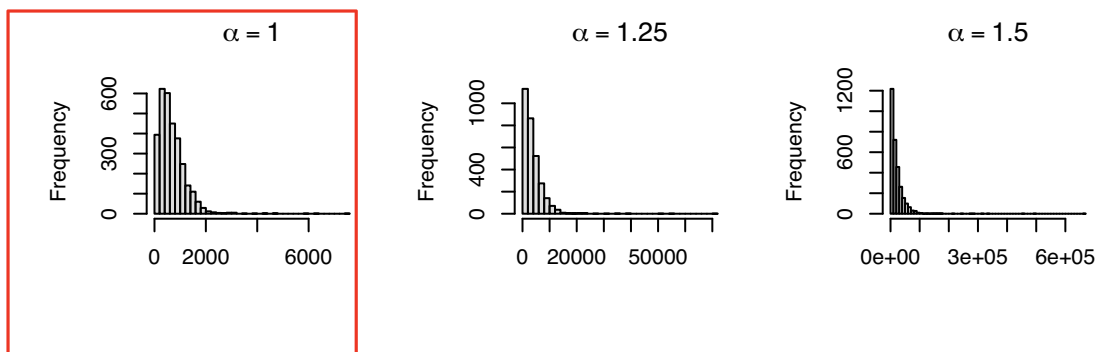
- Varying the power on the number of farms in 1987

```
par(mfrow=c(3,3))
a = seq(-1/2, 1.5, length.out=9)
for (i in 1:9) {
  hist( powerfun(agpop$farms87 + 1, a[i]), col=adjustcolor("grey", alpha = 0.5),
    main= bquote(alpha == .(a[i])), xlab="", breaks=50 )
}
```

Decrease α histogram bump moves right



increase α histogram bump moves left



- Notice how the “bump” changes as α changes.
- Which values of α make the population histogram more symmetric? ($\alpha = 0.25$)
 - Should we find an optimal α ?
- The α values can take any real value in principle,
 - but in practice, we restrict the values to a small set.
 - The powers should be restricted to those which are easily interpretable.
- * John Tukey suggested (Tukey 1977) imagining that the set of powers were arranged in a “ladder” with the smallest powers on the bottom and the largest on the top.

Now one simply moves “up” or “down” on Tukey’s ladder of powers to arrive at a re-expression that achieves the desired effect on the data values.

alpha	ladder
...	up
2	-----
1	-----
1/2	-----
1/3	-----
0	-----
-1/3	-----
-1/2	-----
-1	-----
-2	-----
...	down

How to pick α ?

Two different, but related, effects of transformation are often of interest:

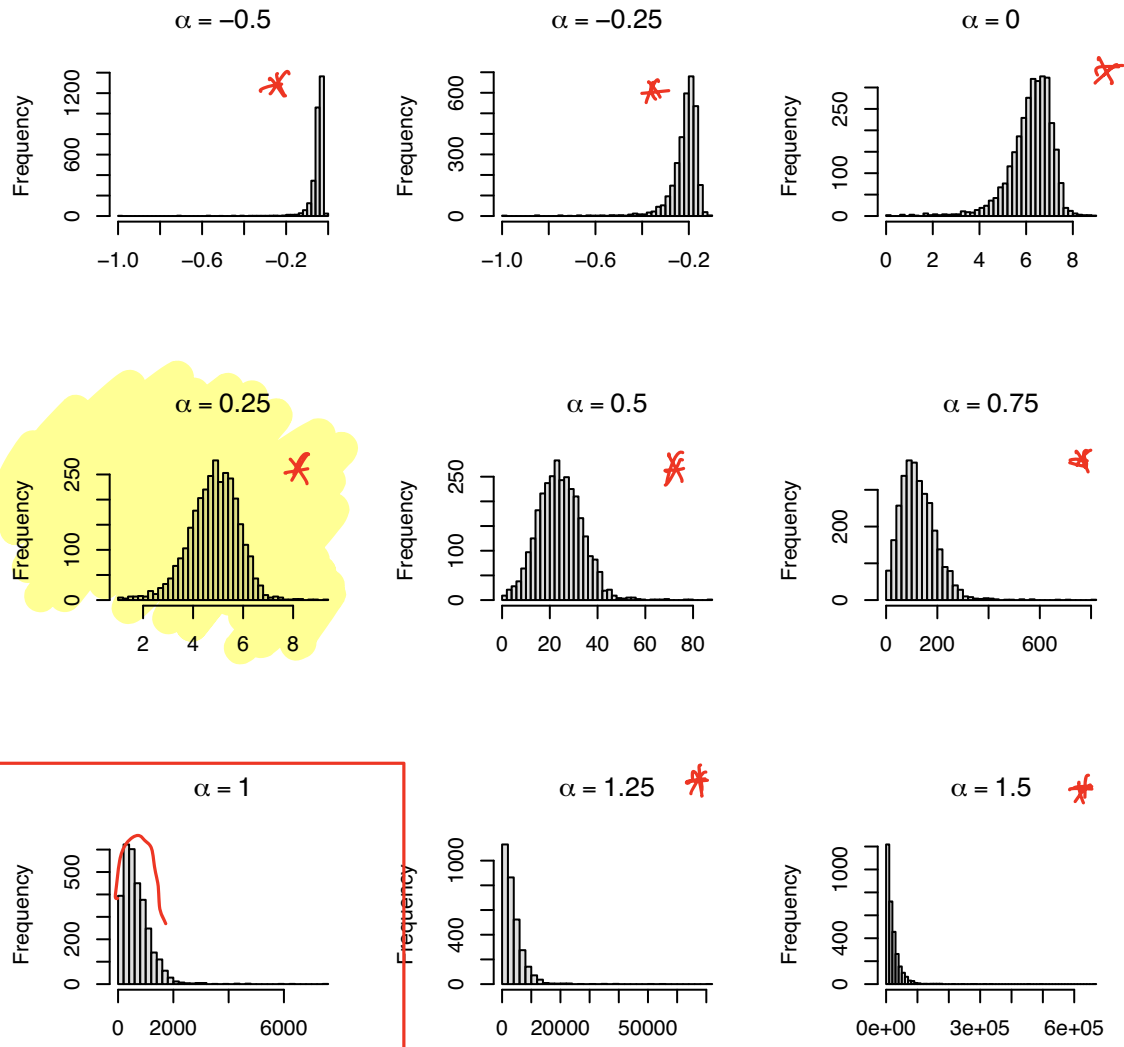
- First, producing a more symmetric looking histogram
- Second, producing roughly linear scatter-plots
 - Imagine (for all $u \in \mathcal{P}$) a scatter-plot of all pairs (x_u, y_u) .
 - Can we change the powers α_x and α_y for each such that the scatter-plot of the re-expressed pairs $(T_{\alpha_x}(x), T_{\alpha_y}(y))$ linearly on a straight line?
- Fortunately, for each of these effects there is a corresponding “bump rule” that indicates the direction (up or down) to move on Tukey’s ladder to achieve it.

Bump Rule 1: Making histograms more symmetric

- The rule is that the location of the “bump” in the histogram (where the points are concentrated) tells you which way to “move” on the ladder.
 - If the bump is on “lower” values, then move the power “lower” on the ladder;
 - If it is on the “higher” values, then move the power “higher” on the ladder.
- Does this rule agree with the 1987 farming data? (re-plotted here)

```
par(mfrow=c(3,3))
a = seq(-1/2, 1.5, length.out=9)

for (i in 1:9) {
  hist( powerfun(agpop$farms87 + 1, a[i]), col=adjustcolor("grey", alpha = 0.5),
        main= bquote(alpha == .(a[i])), xlab="", breaks=50 )
}
```

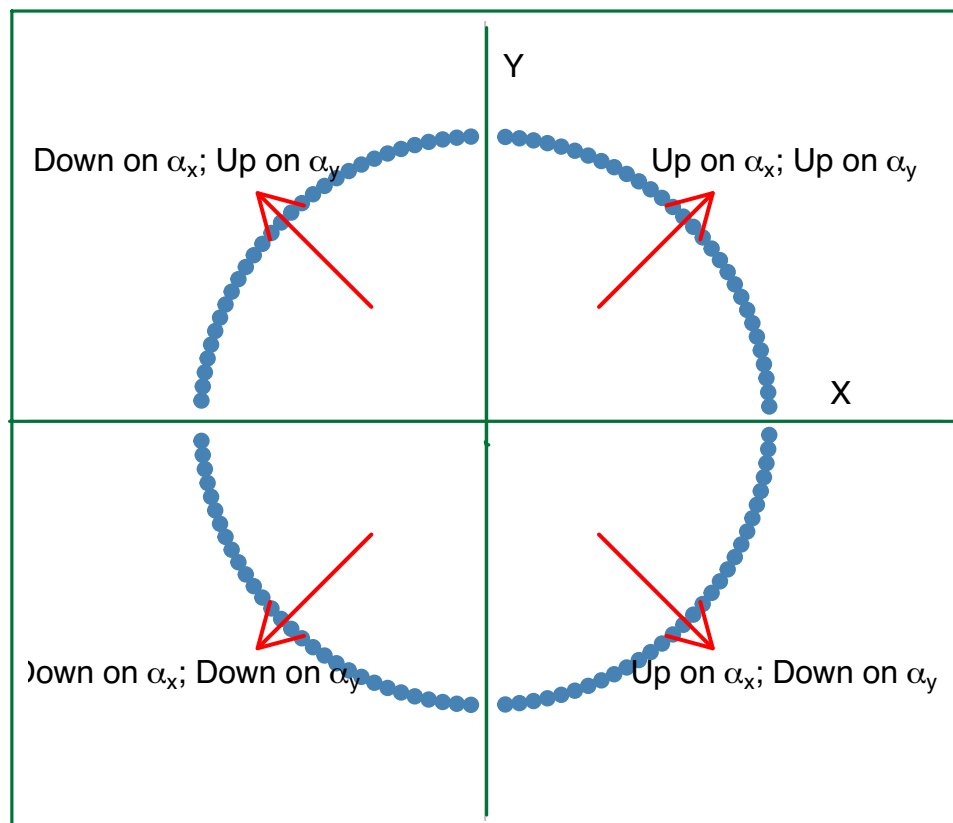


Bump Rule 2: Straightening Scatter-plots

A scatter-plot of (x_u, y_u) for $u \in \mathcal{P}$ may be “straightened” by applying (possibly) different power transforms to each coordinate to give a new (hopefully straighter looking) scatter-plot of the re-expressed data $(T_{\alpha_x}(x_u), T_{\alpha_y}(y_u))$.

- Because each of the coordinates has its own power transformation, there will be two different ladders of transformation
 - the x ladder and
 - the y ladder.
- As with histograms, there is a “bump rule” to tell you how to move on the ladder.
 - ✂ In the case of scatter-plots, the “bump” corresponds to the curvature appearing in the scatter-plot.
 - This is only approximate in practice, but reduces to one of four different possibilities:

Each quadrant shows a monotonic curved relation



Direction of the bump suggests ladder moves

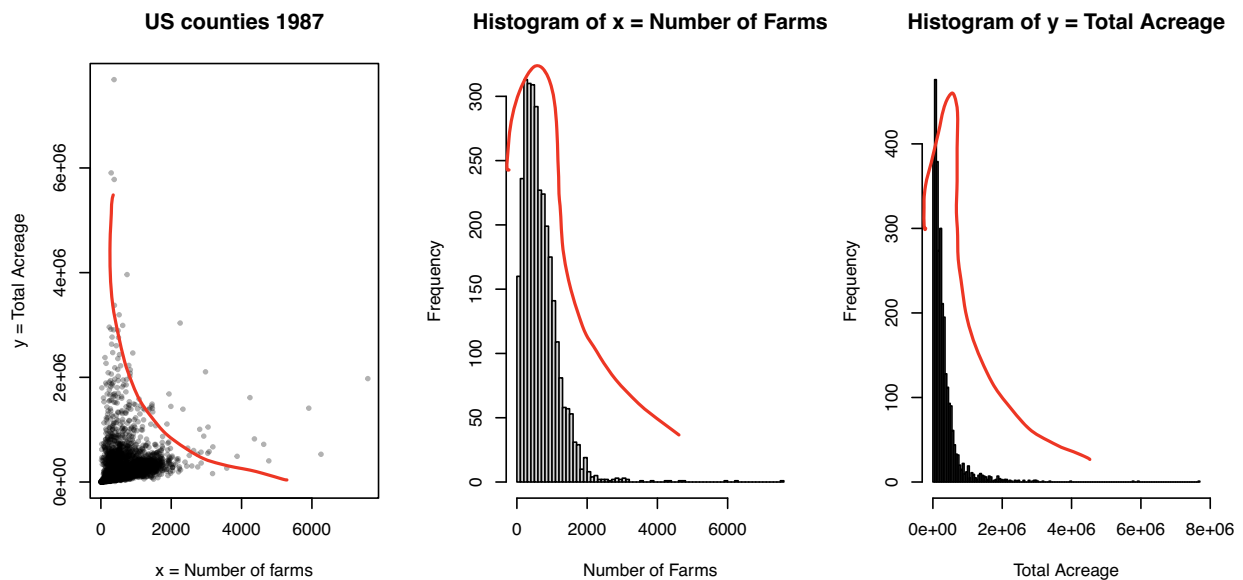
- Example: Agriculture Data

– 1987: Number of Farms vs. Total Acreage of Farming

```
par(mfrow=c(1,3))

plot( agpop$farms87, agpop$acres87, pch = 19, cex=0.5,
      col=adjustcolor("black", alpha = 0.3),
      xlab = "x = Number of farms", ylab = "y = Total Acreage",
      main = "US counties 1987")

hist( agpop$farms87, breaks = "FD",
      xlab="Number of Farms",
      main = "Histogram of x = Number of Farms")
hist( agpop$acres87, breaks = "FD",
      xlab="Total Acreage",
      main = "Histogram of y = Total Acreage")
```



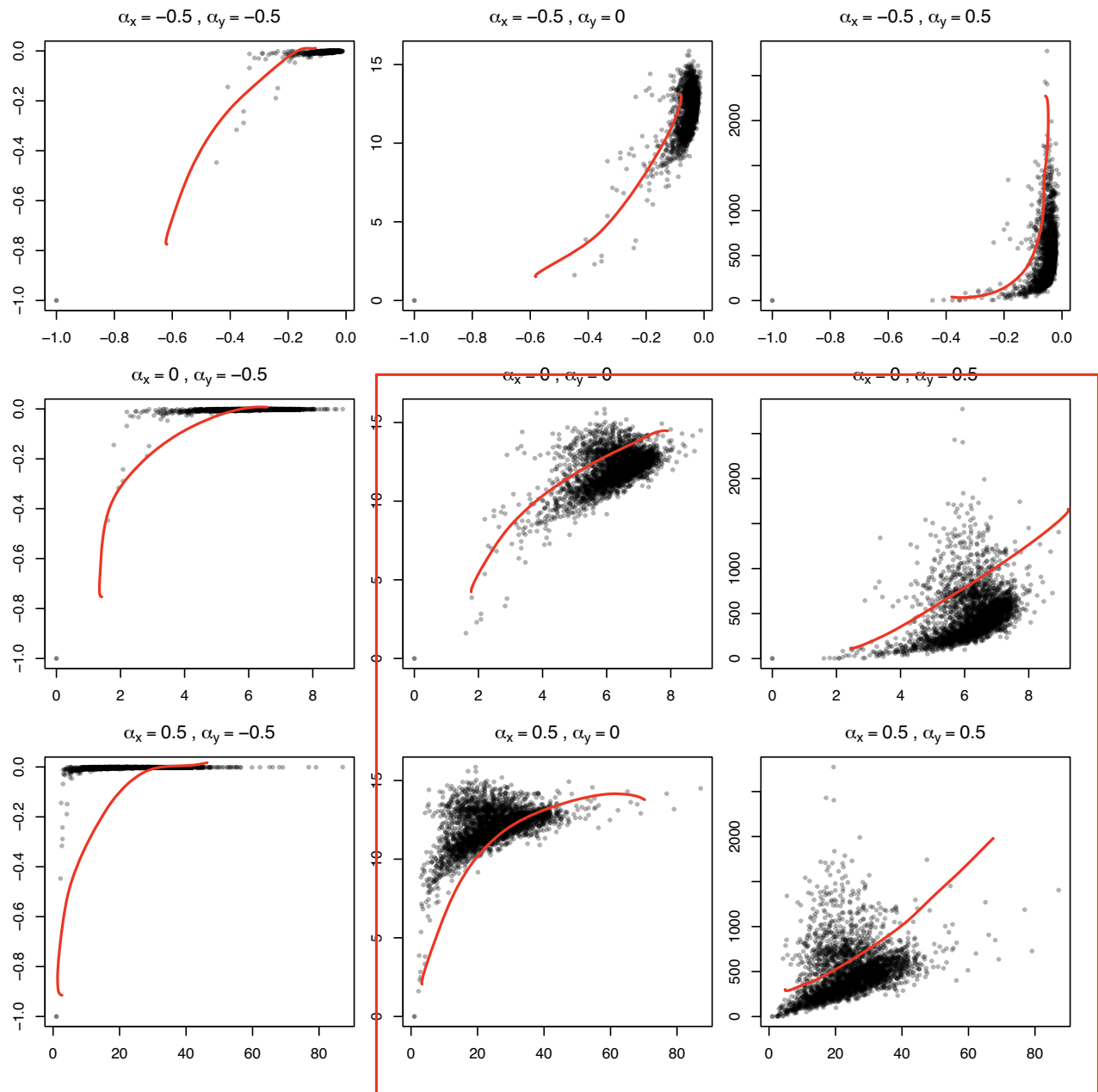
- We can apply a different power transformation to each variate (x = Number of Farms; y = Total Acreage of Farming).

```
par(mfrow=c(3,3), mar=2.5*c(1,1,1,0.1))
a = rep(c(-1/2,0,1/2),each=3) ←  $x$ 
b = rep(c(-1/2,0,1/2),times=3) ←  $y$ 
subdata = agpop[,c('farms87', 'acres87')]
subdata = na.omit(subdata )
```

```

for (i in 1:9) {
plot( powerfun(subdata$farms87+1, a[i]), powerfun(subdata$acres87+1, b[i]), pch = 19, cex=0.5,
      col=adjustcolor("black", alpha = 0.3), xlab = "", ylab = "",
      main = bquote(alpha[x] == .(a[i]) ~ "," ~ alpha[y] == .(b[i])))
}

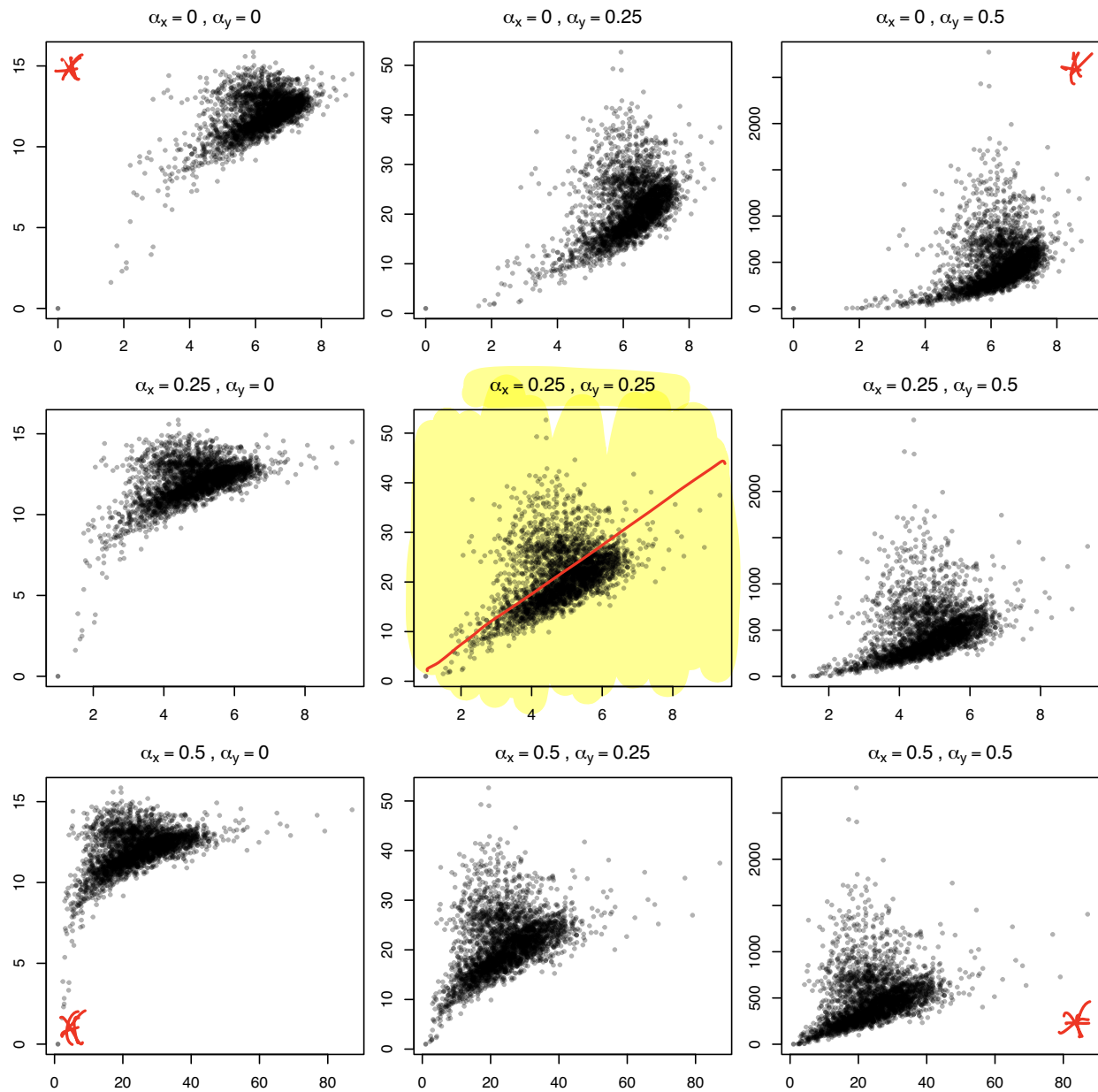
```



- Which α_x and α_y help straighten the relationship?

- Refining the search grid, we have

```
par(mfrow=c(3,3), mar=2.5*c(1,1,1,0.1))
a = rep(c(0,1/4,1/2),each=3)  $\leftarrow \alpha_x$ 
b = rep(c(0,1/4,1/2),times=3)  $\leftarrow \alpha_y$ 
subdata = agpop[,c('farms87', 'acres87')]
subdata = na.omit(subdata)
for (i in 1:9) {
  plot( powerfun(subdata$farms87+1, a[i]), powerfun(subdata$acres87+1, b[i]), pch = 19, cex=0.5,
        col=adjustcolor("black", alpha = 0.3), xlab = "", ylab = "",
        main = bquote(alpha[x] == .(a[i]) ~ "," ~ alpha[y] == .(b[i])))
}
```



Summary

- We examined properties of the power transformation.
- We reviewed two bump rules:
 - The first rule makes histograms (roughly) more symmetric.
 - The second rule makes scatter-plots (roughly) straighter.

Exercises

From the Explicit Attribute exercises try problems 7, 8, 19 and 22. Also, see problem 1 from the Implicit Attributes exercises.

2.2.6 Order, Rank, and Quantiles

Overview

- We examine the relationship between order and rank statistics. 
- We see how to use order and rank statistics to obtain quantiles.
- We discuss several quantile-based location and scale attributes. 
- We investigate the the relationship between quantile plots and histograms. 

Population attributes can also be an indexed collection of values. For example, consider the following different attributes

- Recall the **order statistics**:

$$y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(N)}$$

which are the ordered values (including ties) of the variate values $y_u \in \mathcal{P}$

- The **rank statistics**:

$$r_1, r_2, \dots, r_N$$

which are the **ranks** of the variate values $y_1, y_2, \dots, y_N$ from the $y_u \in \mathcal{P}$.

- For example, if $y_i = y_{(k)}$ then y_i is the k^{th} smallest value and so y_i has rank $r_i = k$. This means that

$$y_{(r_u)} = y_u \quad \forall u \in \mathcal{P}$$

Order and Rank: Example

A small population with five units and a single variate y :

```
y <- c(22, 3, 12, 1, 42)
y
```

```
## [1] 22 3 12 1 42
```

The order statistics:

```
y.ordered <- sort(y)
y.ordered
```

```
## [1] 1 3 12 22 42  $y_{(1)}, y_{(2)}, y_{(3)}, y_{(4)}, y_{(5)}$ 
```

The rank statistics (Note, no ties to worry about):

```
y.rank <- rank(y)
y.rank
```

```
## [1] 4 2 3 1 5  $r_1, r_2, r_3, r_4, r_5$ 
```

The connection between them:

```
y.ordered[y.rank] == y
```

```
## [1] TRUE TRUE TRUE TRUE TRUE
```

Scatter-plot for y_u vs. rank

These two attributes are often combined as a single graphical attribute by plotting the pairs (r_u, y_u) or, equivalently, $(u, y_{(u)})$ for all $u \in \mathcal{P}$.

- For example, the variate acres87 from the agricultural census data:

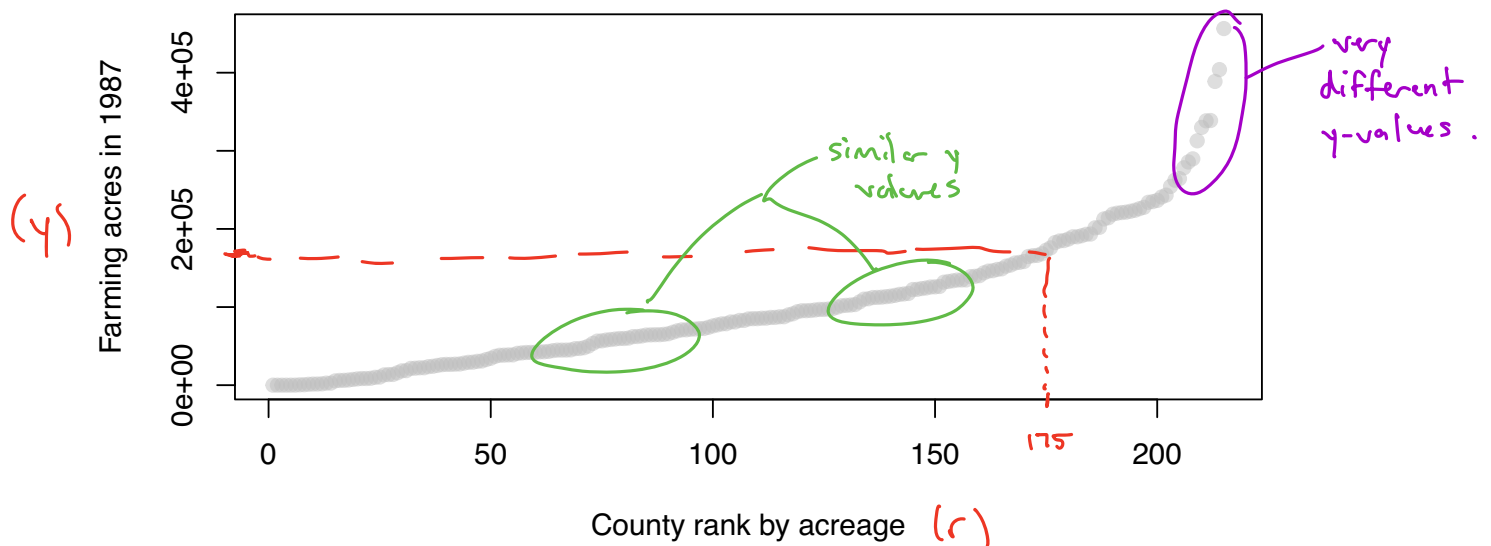
```
directory <- "../..Data"
dirsep <- "/"
filename <- paste(directory, "agpop_data.csv", sep=dirsep)
agpop <- read.csv(filename, header=TRUE)
agpop[agpop == -99] = NA
```

```
### read the data from wherever it is stored, e.g. in some directory named Data
```

```
→ y <- agpop$acres87[agpop$region == "NE"]
y <- na.omit(y)
yrank <- rank(y, ties.method = "first") # Now ensure ties appear in data set order
```

```
plot( $r_u$  yrank,  $y_u$  y, pch = 19, col=adjustcolor("grey", alpha = 0.5),
     xlab = "County rank by acreage",
     ylab = "Farming acres in 1987",
     main = "Counties in the North East USA")
```

Counties in the North East USA



Notes

- the height at any point tells the location of the value of y
- horizontal location identifies where in the order of the variate values that unit appears
- the plot is monotonically non-decreasing from left to right.
- flat spots indicate tied values of y ; nearly flat spots are counties where the number of acres under farming are nearly the same.
- rapidly rising spots are counties which, though near each other in order (rank), are very different in the actual values of y (acreage).

Quantiles

* Rather than using ranks, it can be more convenient to use the proportion of units in the population having a value less-than-or-equal-to y .

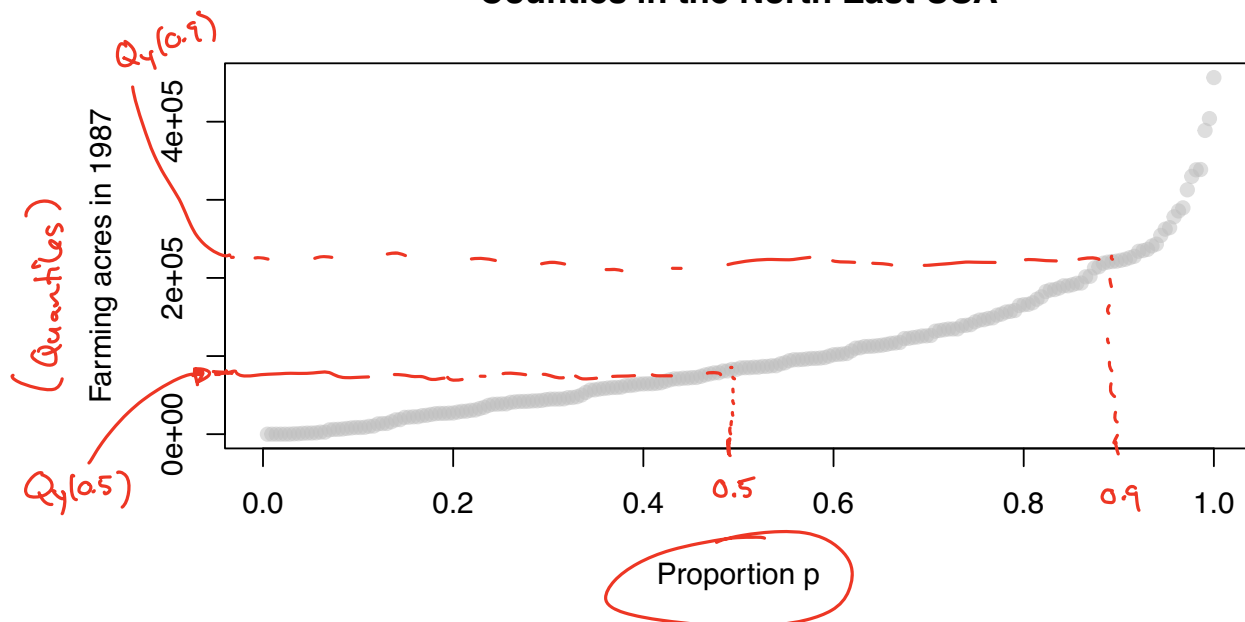
- So instead of plotting the pairs (r_u, y_u) , we could equivalently plot the pairs (p_u, y_u) where

$$p_u = \frac{r_u}{N}$$

is the proportion of the units $i \in \mathcal{P}$ whose value $y_i \leq y_u$

```
N <- length(y)
p <- yrank/N
plot(p, y, pch = 19, col=adjustcolor("grey", alpha = 0.5),
     xlim=c(0,1),
     xlab = "Proportion p",
     ylab = "Farming acres in 1987",
     main = "Counties in the North East USA")
```

Counties in the North East USA



• Notes

- The middle value or proportion equal to $1/2$ corresponds to the median.
- The values on the y -axis are the quantiles.

- Strictly speaking, the plotted points are $(p, Q_y(p))$ where

- $p \in \{\frac{1}{N}, \frac{2}{N}, \dots, 1\}$ and

- $Q_y(p)$ is the p^{th} **quantile** of y

$$Q_y(p) = y_{(N \times p)}$$

- and is sometimes called the **quantile function** of y for all $p \in [\frac{1}{N}, 1]$.

- The quantile function is a population attribute which can be used to generate a number of other interesting population attributes:

- the quantile $Q_y(p)$ for any p locates the variate values in the population, and is thus a **measure of location**.
- most (but not all) location measures try to capture **central** tendency.

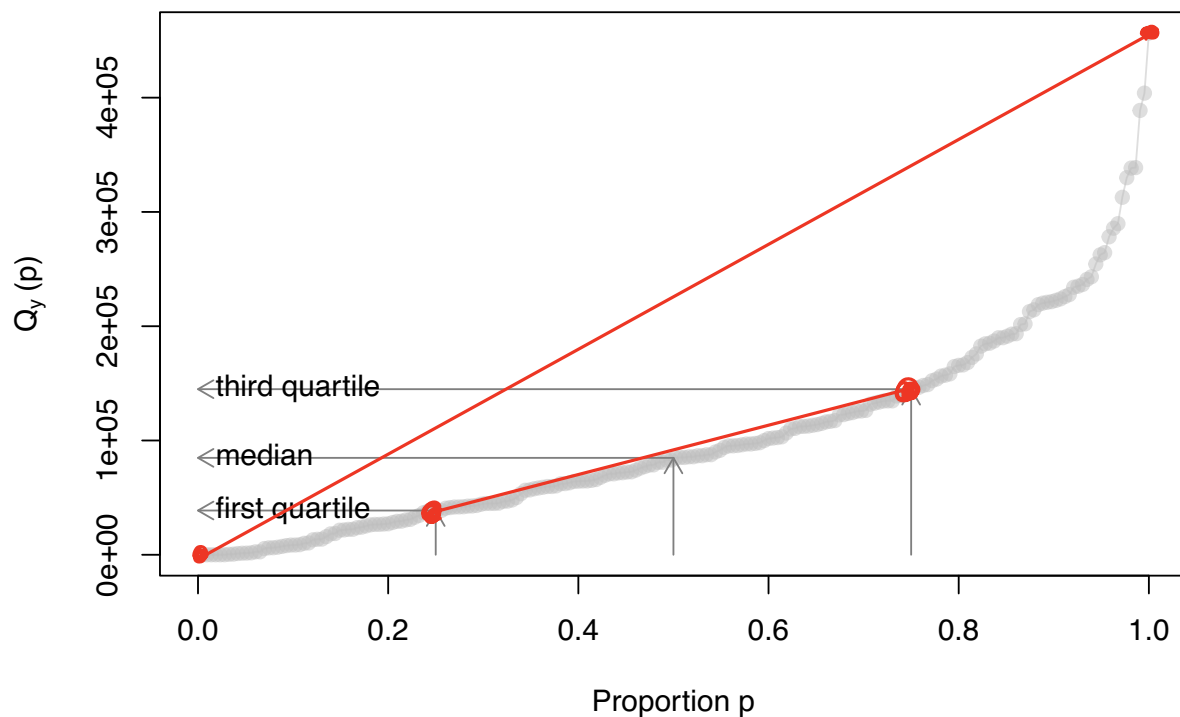
Quantiles that measure center

- the **median**: $Q_y(1/2)$
- the **mid-hinge** (average of the first and third quartiles): $\frac{Q_y(1/4)+Q_y(3/4)}{2}$
- the **mid-range** (average of the minimum and maximum): $\frac{Q_y(1/N)+Q_y(1)}{2}$
- the **trimean**: $\frac{Q_y(1/4)+2 \times Q_y(1/2)+Q_y(3/4)}{4}$

These can be readily obtained from the quantile plot.

- Reading off the vertical location of $Q_y(p)$ for any pre-determined p provides some measure of location.

1987 farming acreage for north east counties



Quantiles that measure spread

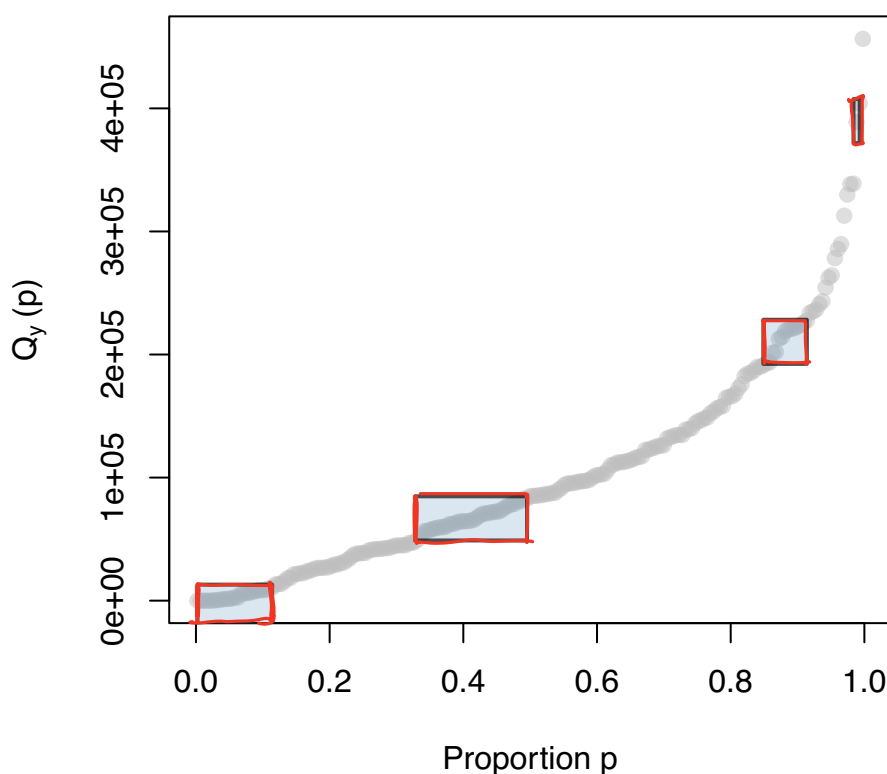
- The quantile function can also be used to provide some natural measures of **spread** for the variate y :
 - the **range**: $Q_y(1) - Q_y(1/N)$
 - the **inter-quartile range**: $IQR_y = Q_y(3/4) - Q_y(1/4)$
 - the **central $100 \times p\%$ range**
- Alternatively, the difference between any two quantiles might be divided by the difference in the corresponding p values.
 - That is, the **slope** of the line segment joining any two points $(p_1, Q_y(p_1))$ and $(p_2, Q_y(p_2))$ for $p_1 < p_2$ provides a measure of spread.

Concentration in Quantile Plots

Flatter regions in a quantile plot indicate areas where the variate values appear to be concentrated.

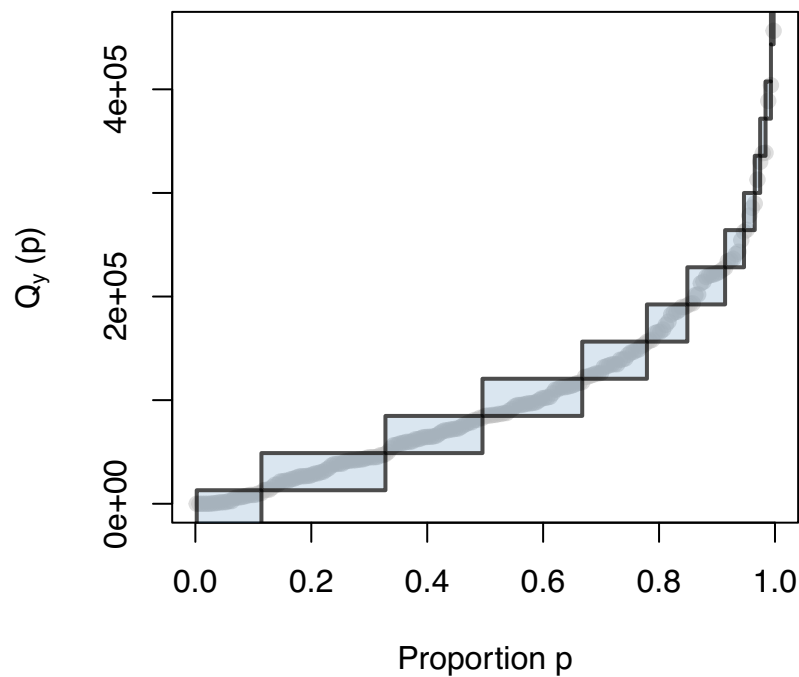
- To quantify this we could draw a box with **fixed height** and see how many elements are within the box.

1987 farming acreage for north east counties



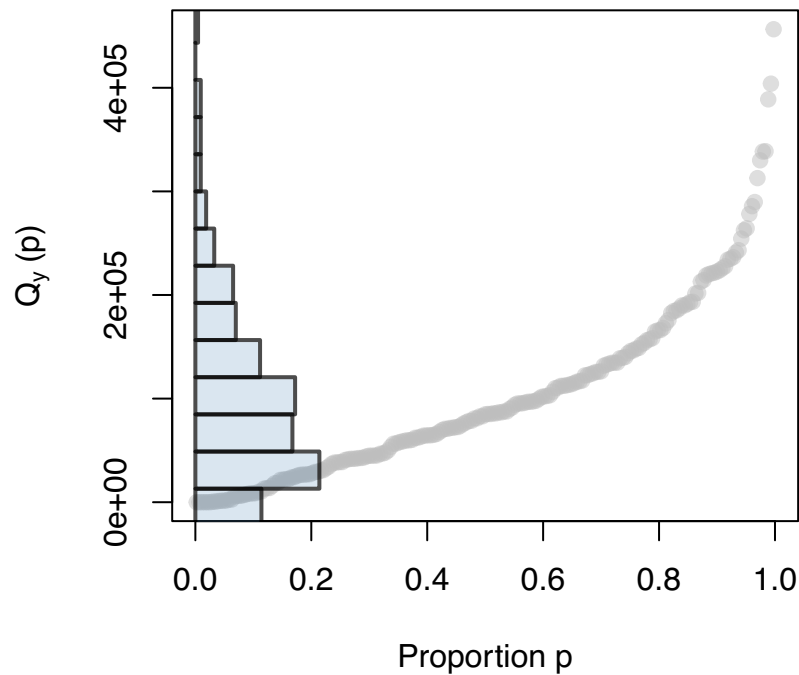
- The width of the box is proportional to the number of elements it contains.
 - The greater the width, the greater the concentration.
- We can produce all such boxes, with **fixed height**, to see how the concentration changes with p .

1987 farming acreage for north east counties



- So how do we interpret these boxes?
 - What happens if we move them all to the left edge of the plot?

1987 farming acreage for north east counties



✱ A histogram of the acreage (or any y variate) is formed from the boxes that identify concentrations on the quantile plot!

Summary

- We examined rank and order statistics.
- We looked at how we can use the quantile plot to summarize a population.
- We reviewed the relationship between histograms and quantile plots.

Exercises

From the Explicit Attribute exercises try problems 13 and 25. For continued learning try problems 15 and 21.